Microsoft Fabric in a nutshell

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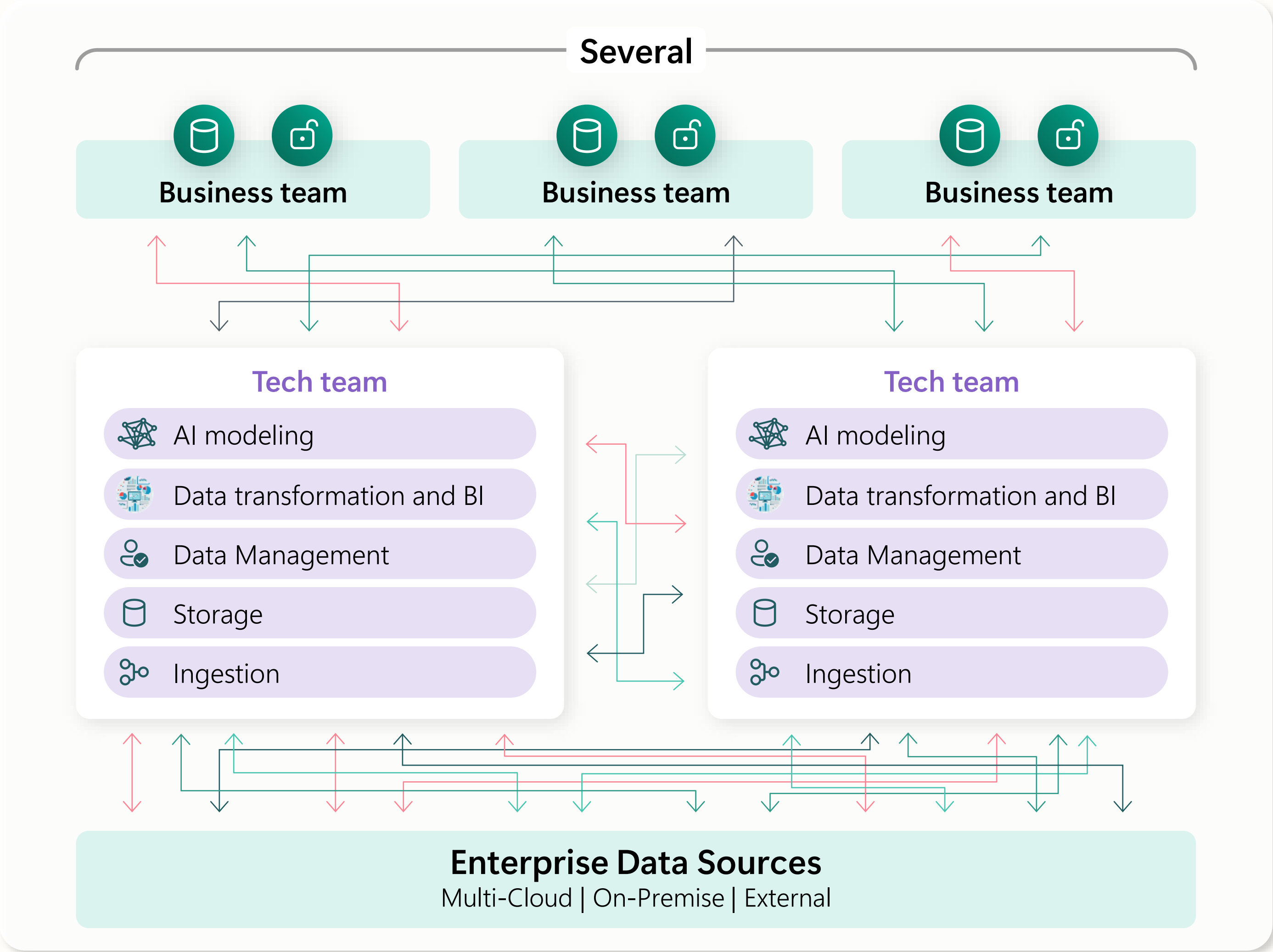
"I am the Chief Information Officer and don't want to be the Chief Integration Officer. Help me translate AI to my competitive advantage."

Every CIO, Every Enterprise

The starting line: A complex, organically evolved data estate

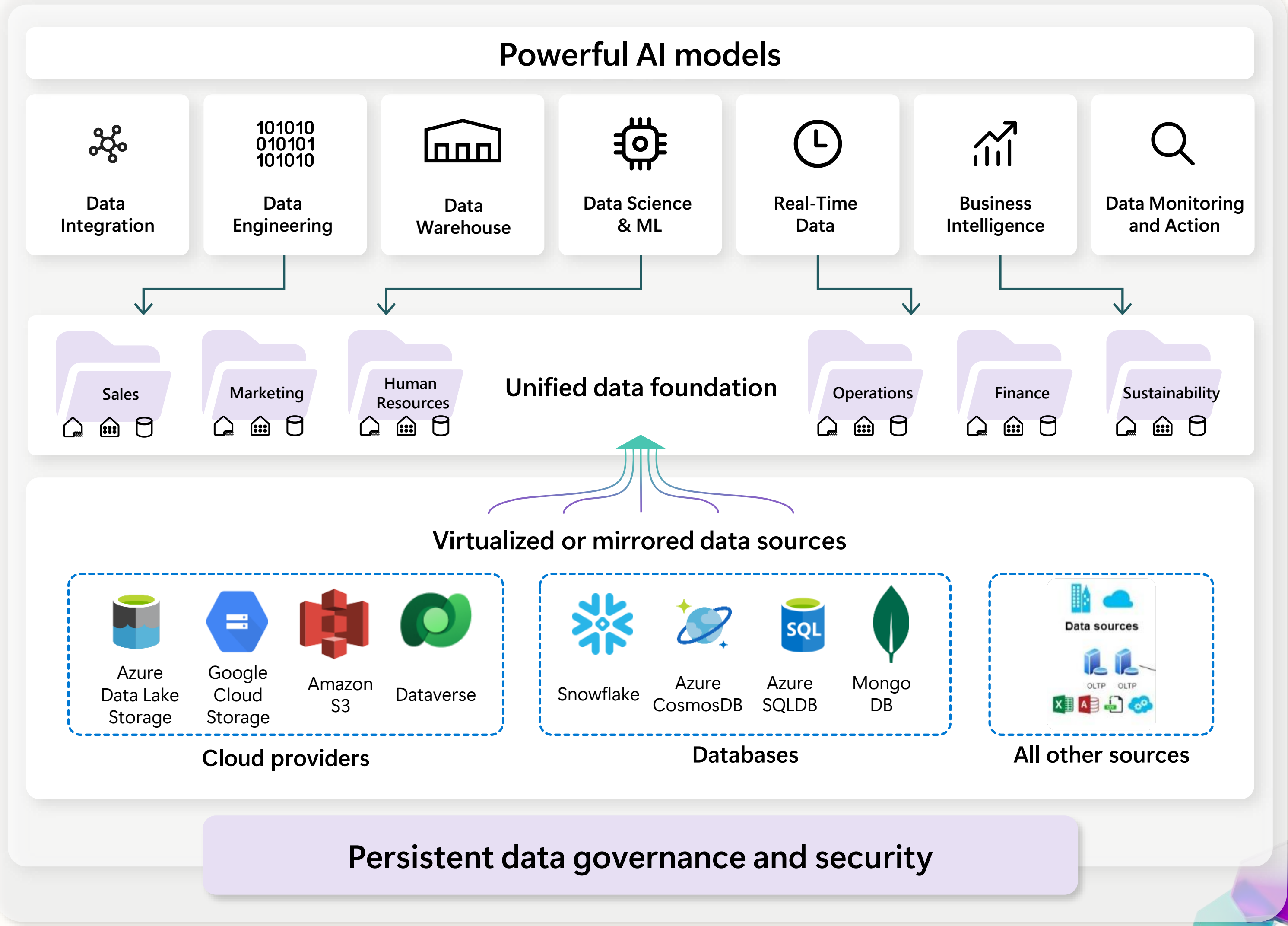


1. Data copies and infrastructure inefficiencies

2. Limited interoperability

3. Data exposure risks

The finish line: A unified, intelligent data estate built on an open and governed data lake house



Microsoft Fabric

The unified data platform for AI transformation

- Data Factory
- Analytics
- Databases
- Real-Time Intelligence
- Power BI

Fabric Platform

- AI
- OneLake
- Governance

Data Factory

Real-Time Intelligence

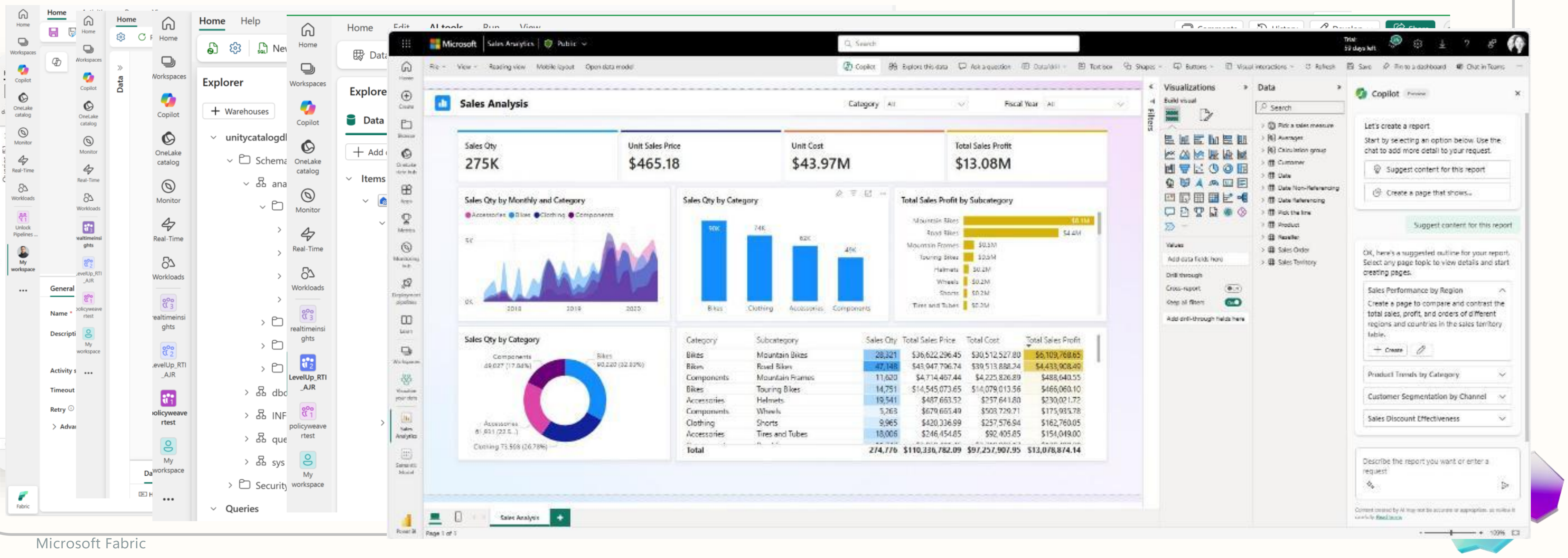
Databases

Data Engineering

Data Science

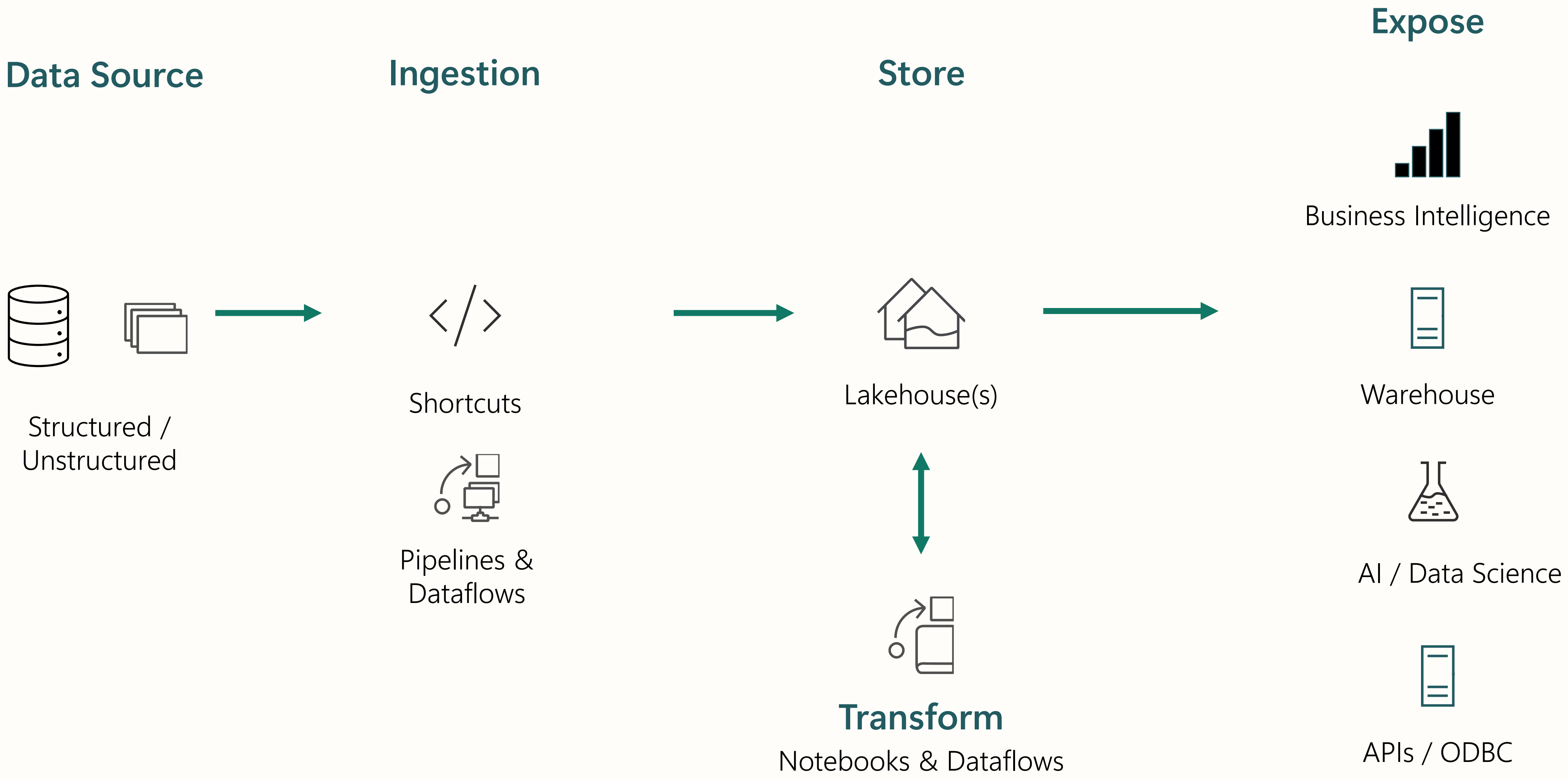
Data Warehouse

Power BI



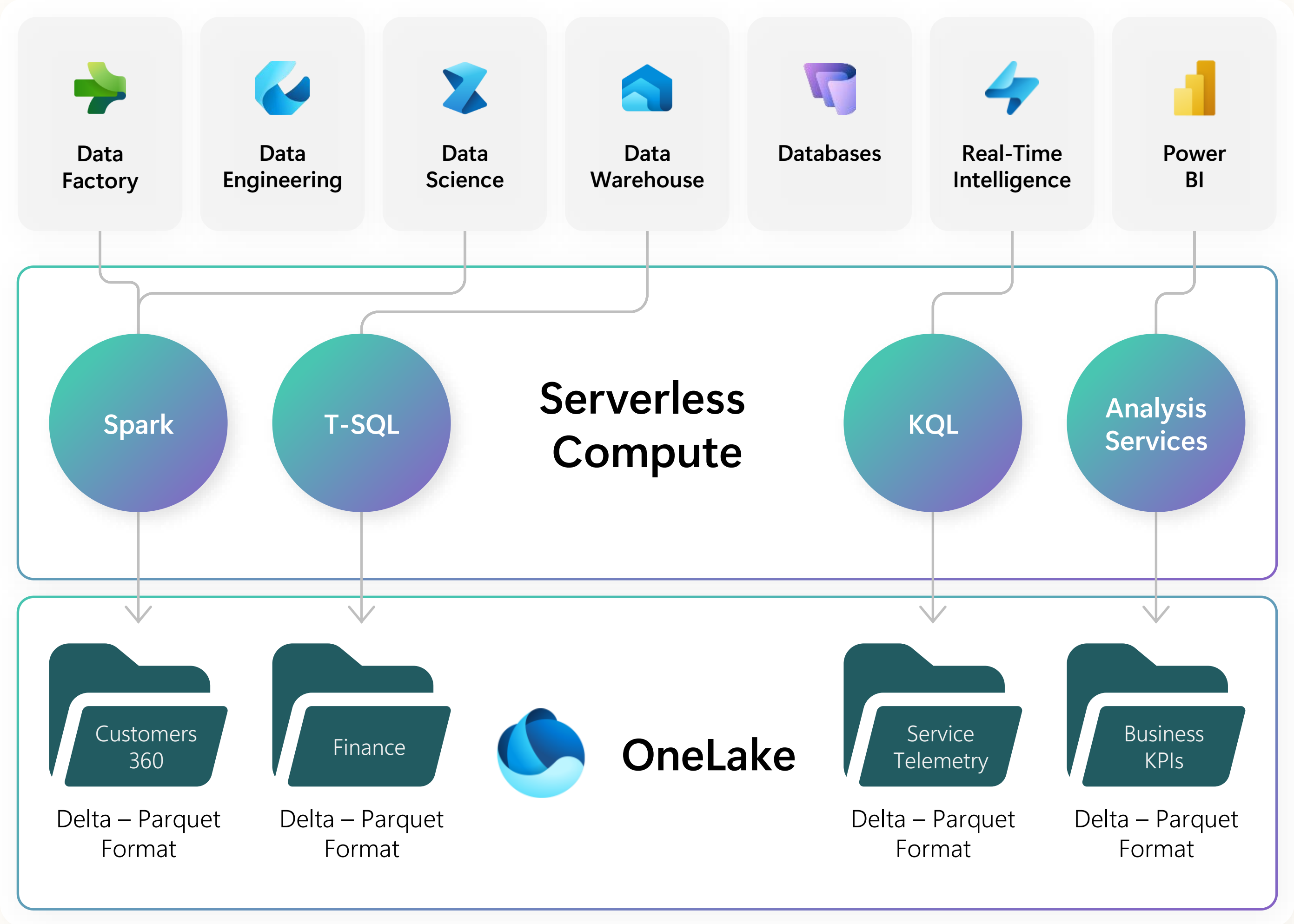
Fabric covers all components of an end-to-end analytics scenario

E.g. classic lakehouse/warehouse workloads



Open Delta Parquet format

Real separation of compute and storage



All the compute engines store their data automatically in OneLake as data items.

The data is stored in a single common format.

Delta – Parquet, an open standards format, and it is the storage format for all tabular data in Fabric.

All the compute engines have been fully optimized to work with Delta Parquet as their native format.

OneLake which logically spans the world

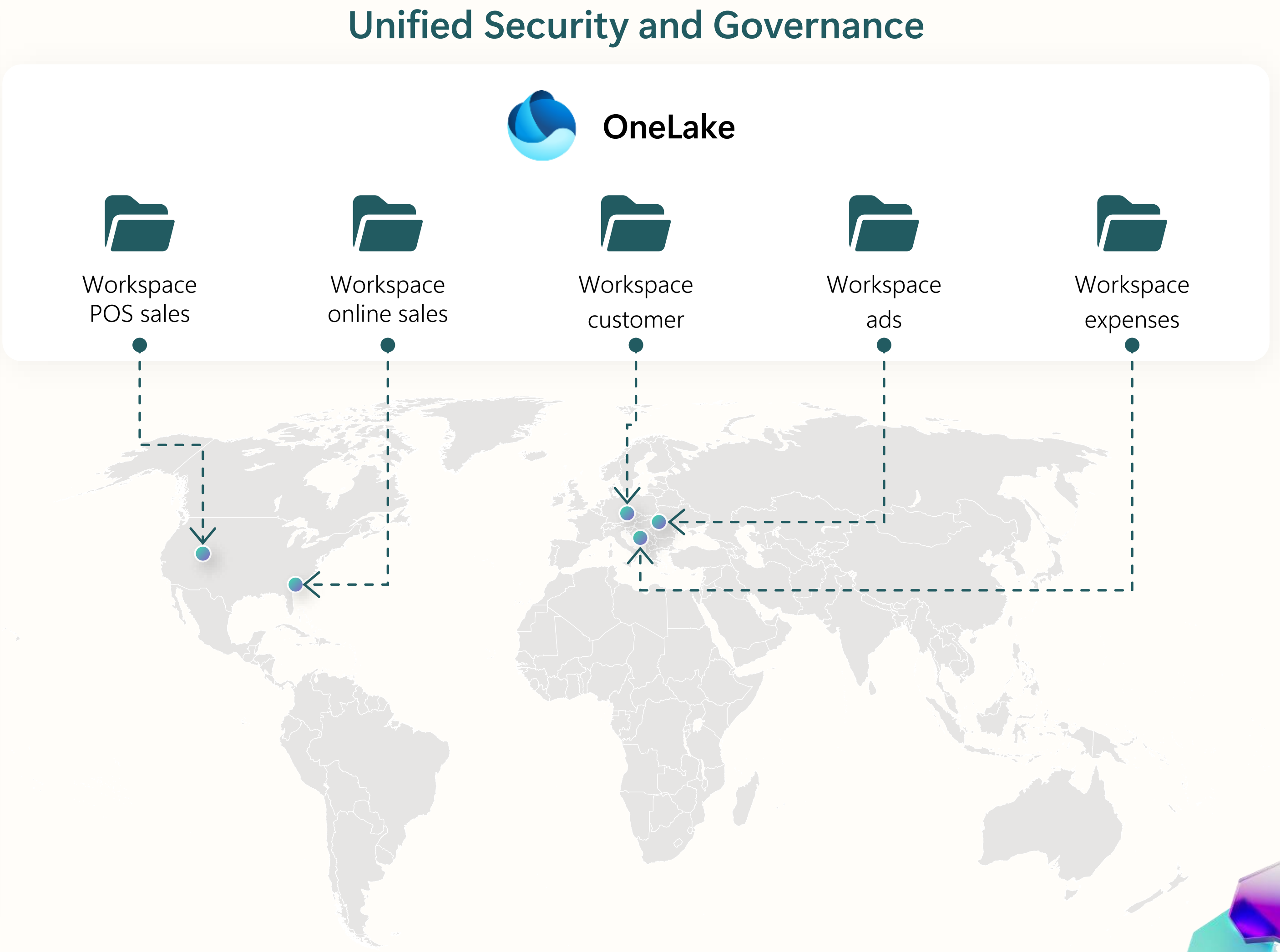
Reside workspaces in different regions around the world while still in the same data lake.

Reside data in different regions without managing different storage resources.

Meet demand for any scale with dedicated storage resources for each workspace (capacity, throughput, and IOPS).

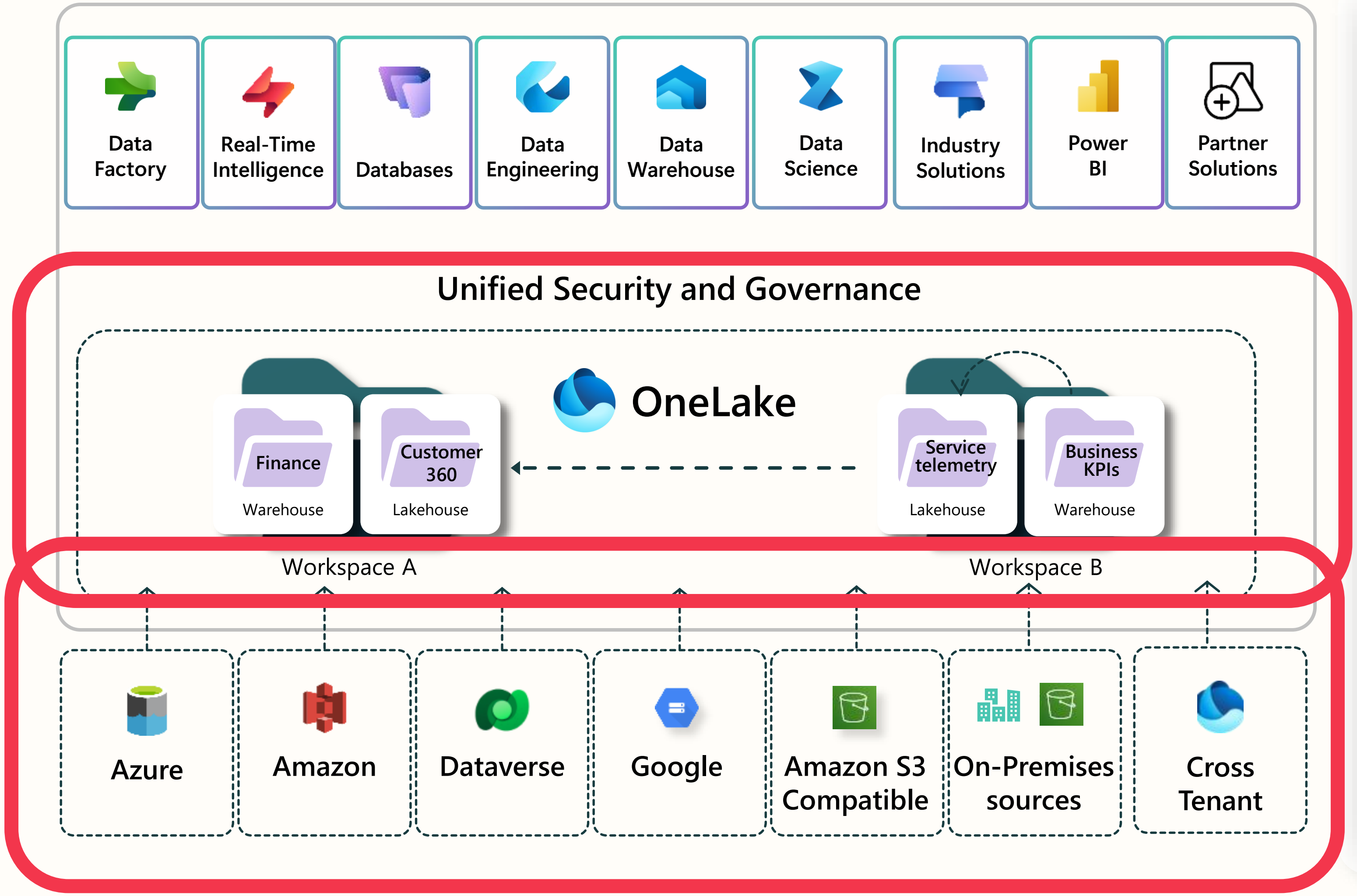
Underlying physical storage is virtualized away.

All storage is zone redundant by default with an option for Geo redundancy.



Shortcuts virtualize data across domains and clouds

No data movements or duplication



A shortcut is a symbolic link which points from one data location to another.

Create a shortcut to make data from a warehouse part of your lakehouse.

Create a shortcut within Fabric to consolidate data across items or workspaces without changing the ownership of the data. Data can be reused multiple times without data duplication.

Existing ADLS Gen2 storage accounts and Amazon S3 buckets can be managed externally to Fabric and Microsoft while still being virtualized into OneLake with shortcuts.

All data is mapped to a unified namespace and can be accessed using the same APIs including the ADLS Gen2 DFS APIs.

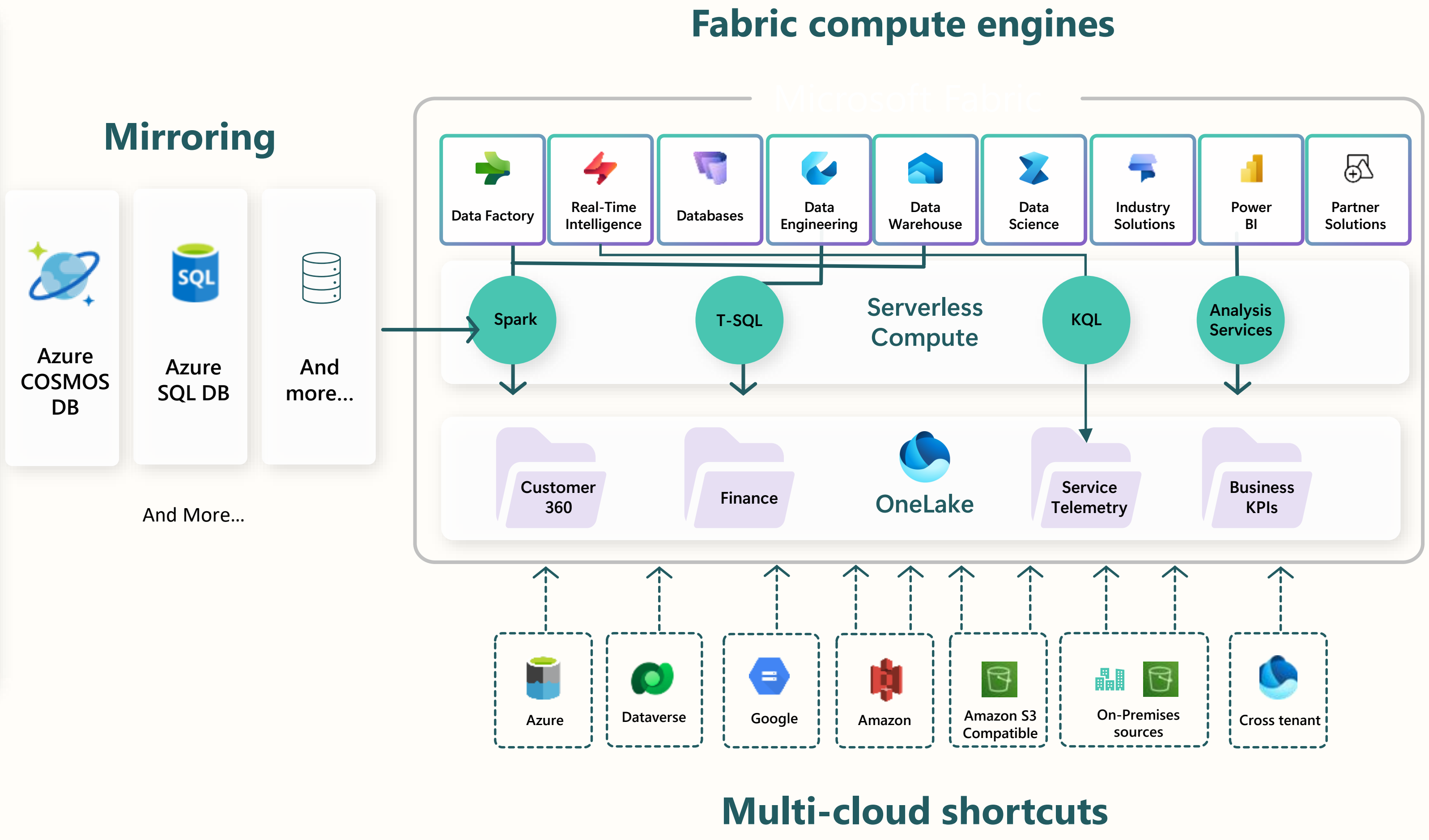
Database Mirroring

Bring data from anywhere

A mirrored item reflects a full external database in OneLake

Ensures the entire database shows up in OneLake, continuously reflecting all changes to the schema and data

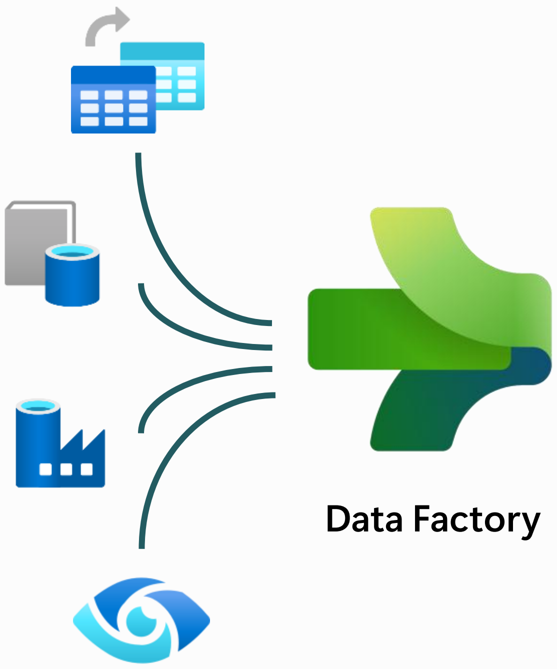
Mirroring can be done either through CDC based replication when the source is store in proprietary format, or by automatically maintaining a set of shortcuts to the external database when the data is stored in open table format (delta and iceberg)



Unifying data in OneLake

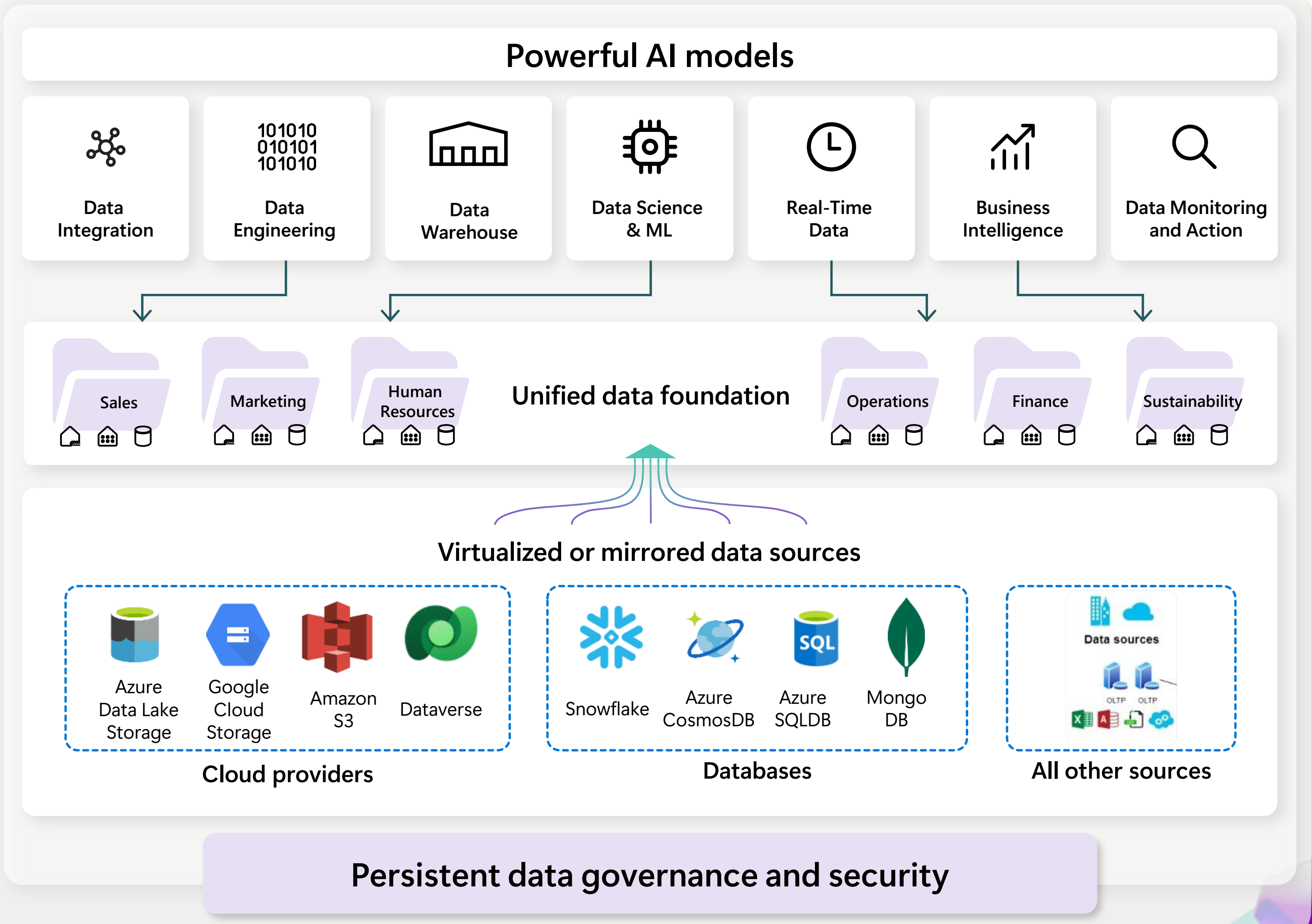
Data Factory

Seamlessly connect to more than 170+ data stores



Azure Database for PostgreSQL	Azure Databricks Delta Lake	Amazon RDS for Oracle	Amazon RDS for SQL Server	Amazon Redshift	Phoenix	PostgreSQL	Presto	Magento (Preview)
Azure SQL Database	Azure SQL Database Managed Instance	Apache Impala	Azure SQL Database Managed Instance	DB2	SAP BW Open Hub	SAP BW via MDX	SAP HANA	Oracle Eloqua (Preview)
Azure Table Storage	MongoDB Atlas	Drill	Google AdWords	Google BigQuery	SAP Table	SQL server	Spark	PayPal (Preview)
Azure Cosmos DB (MongoDB API)	Azure Cosmos DB (SQLAPI)	Greenplum	HBase	Hive	Amazon S3	Amazon S3 Compatible	FTP	SAP Cloud For Customer
Azure Data Lake Storage Gen1	Azure Data Lake Storage Gen1 for Cosmos Structured Stream	Informix	MariaDB	Microsoft Access	File system	Google Cloud Storage (S3APD)	HDFS	Salesforce Marketing Cloud
Azure Data Lake Storage Gen1 for Cosmos Structured Stream	Azure Database for MariaDB	MySQL	Netezza	Oracle	HTTP	Oracle Cloud Storage (S3AP)	SFTP	Shopify (Preview)
Teradata	Vertica	ODBC	OData	REST	Amazon Marketplace Web Service	Concur (Preview)	Dataverse (Common Data Service for App)	Web Table
Jira	Kusto	SharePoint Online List	Dynamics 365	Dynamics AX	Dynamics CRM	Cassandra	Couchbase (Preview)	MongoDB

The finish line: A unified, intelligent data estate built on an open and governed data lake house



Microsoft Fabric

The data platform for the era of AI

AI-powered data platform

Give your data teams all the AI-powered tools they need for any project in an optimized SaaS environment

Open and AI-ready data lake

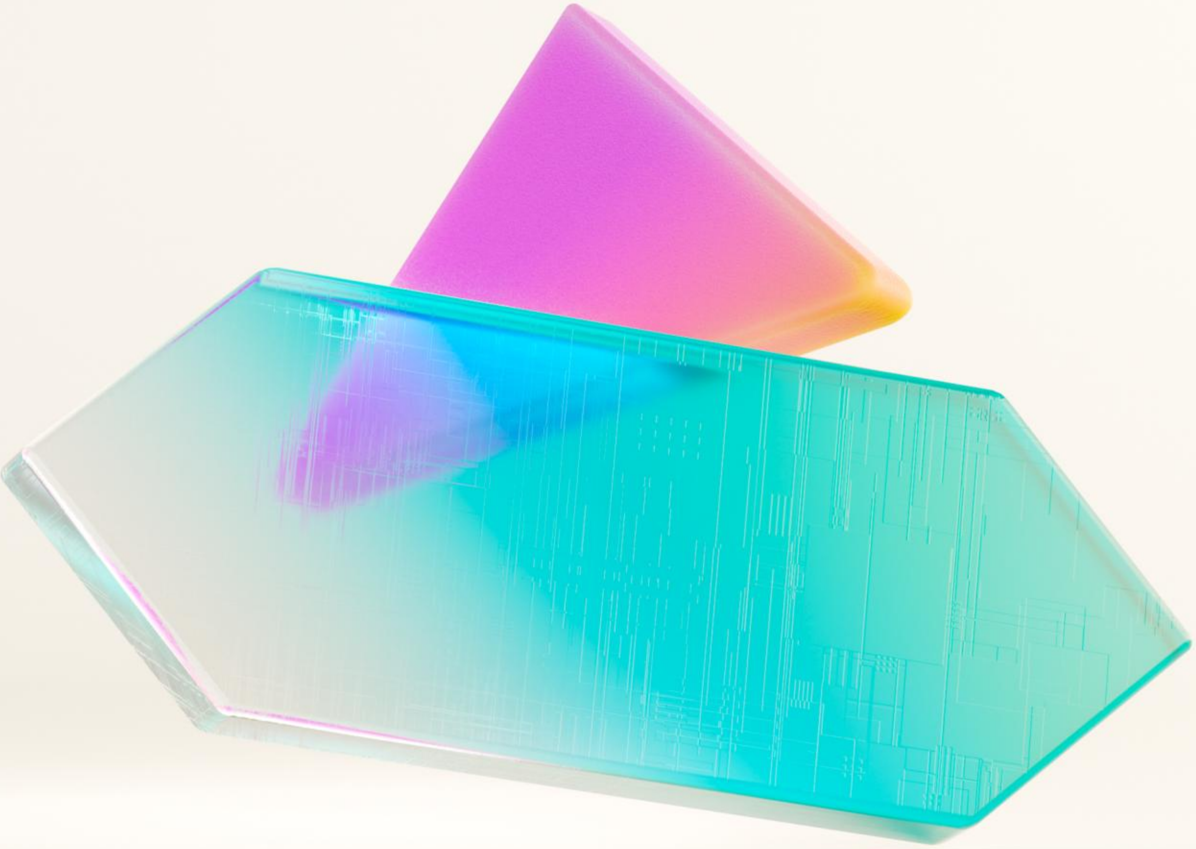
Access your entire multi-cloud data estate from a single data lake and ensure data is ready to power AI innovation

AI-enabled business users

Empower everyone to better understand data with AI-enhanced experiences and visuals embedded in the apps they use everyday

Mission critical foundation

Deploy and manage Fabric with category-leading performance, instant scalability, shared resilience, and built-in security, governance, and compliance



Secure and govern your Environment

Take full control of your data and ensure the right access to the right people

Establish effective federated data security

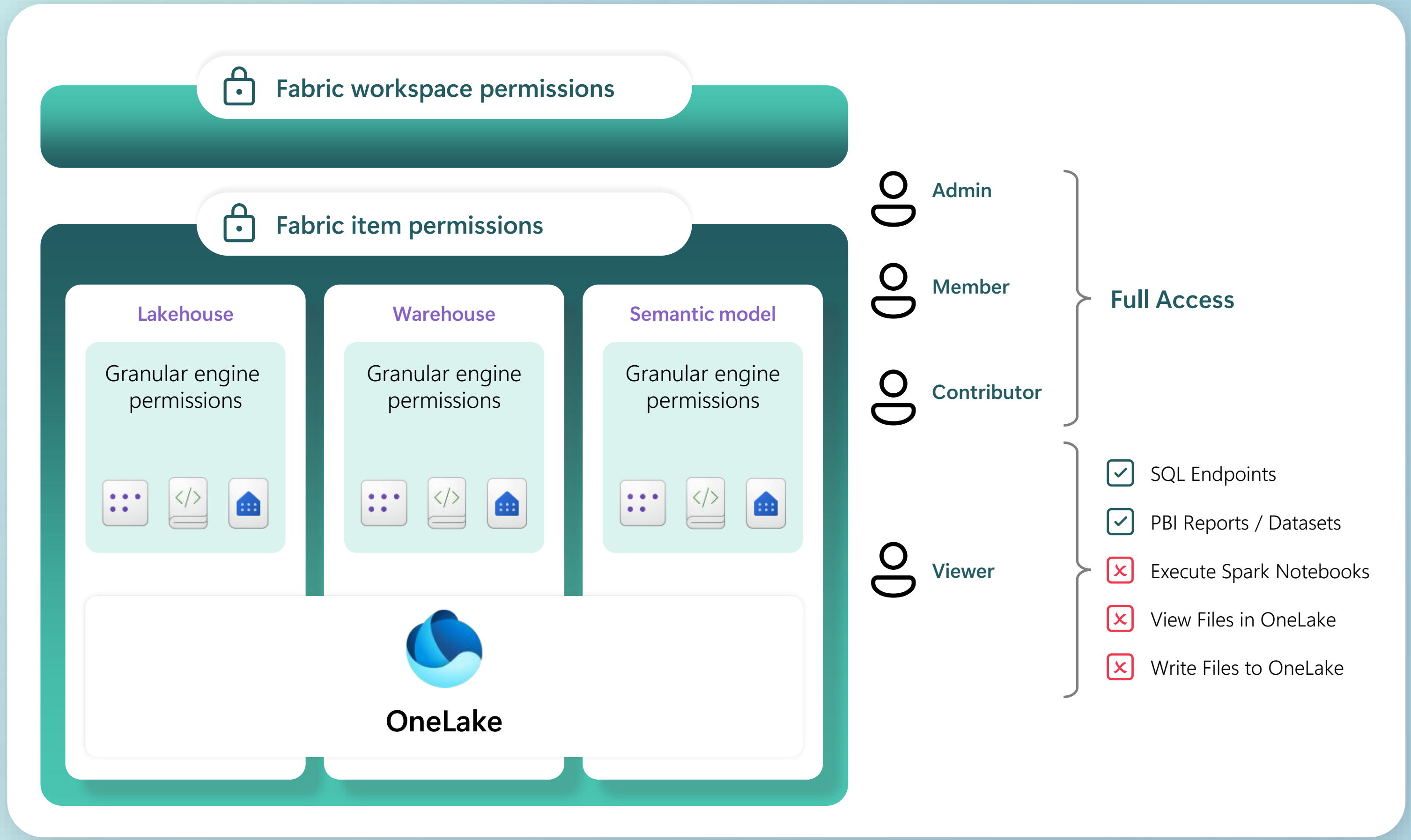
with easy-to-use controls for admins and granular tools for end users

Define security and manage roles at every layer

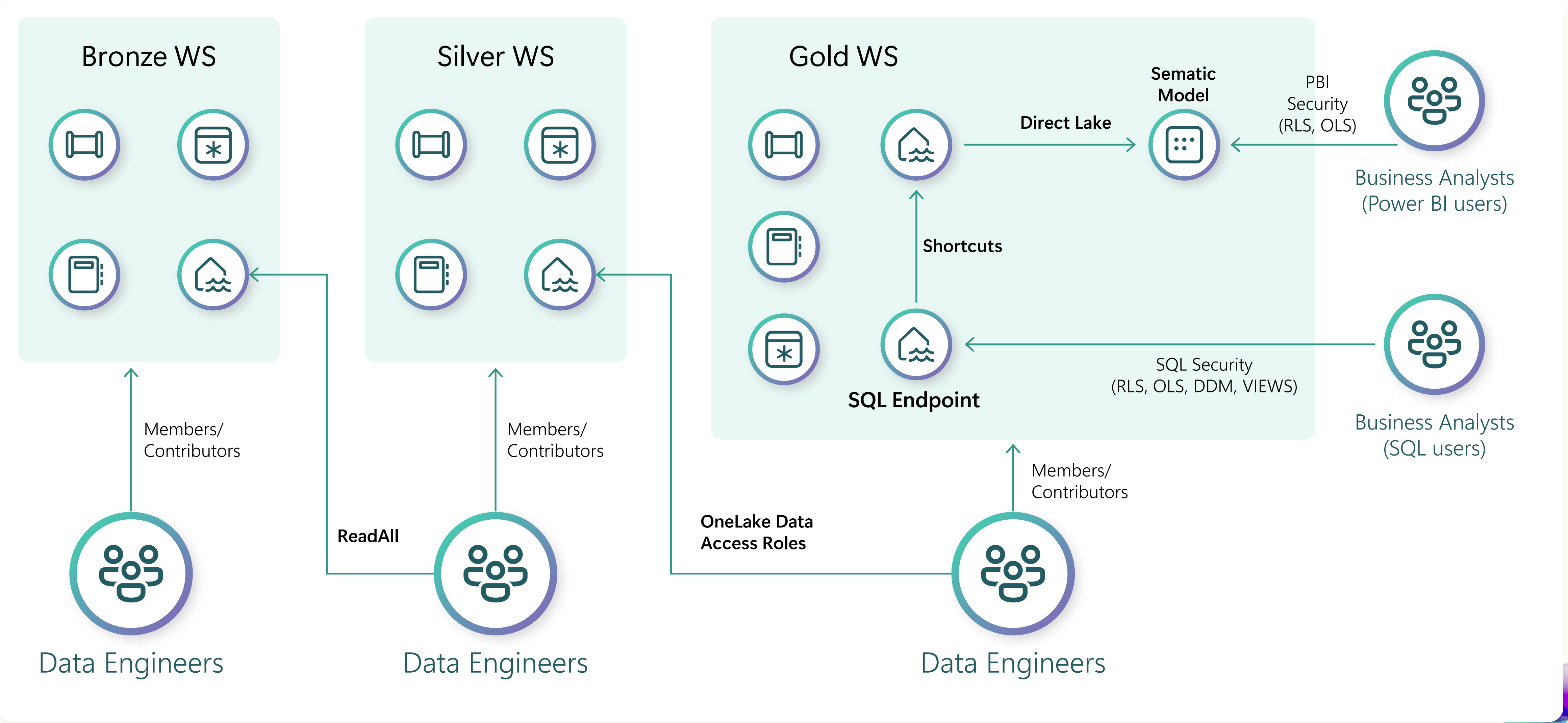
and uniformly enforce access across all Fabric engines so users only access the data they need

Classify sensitive data

with Purview labels, automatically classifying all downstream items and enforcing protection even when the data is exported

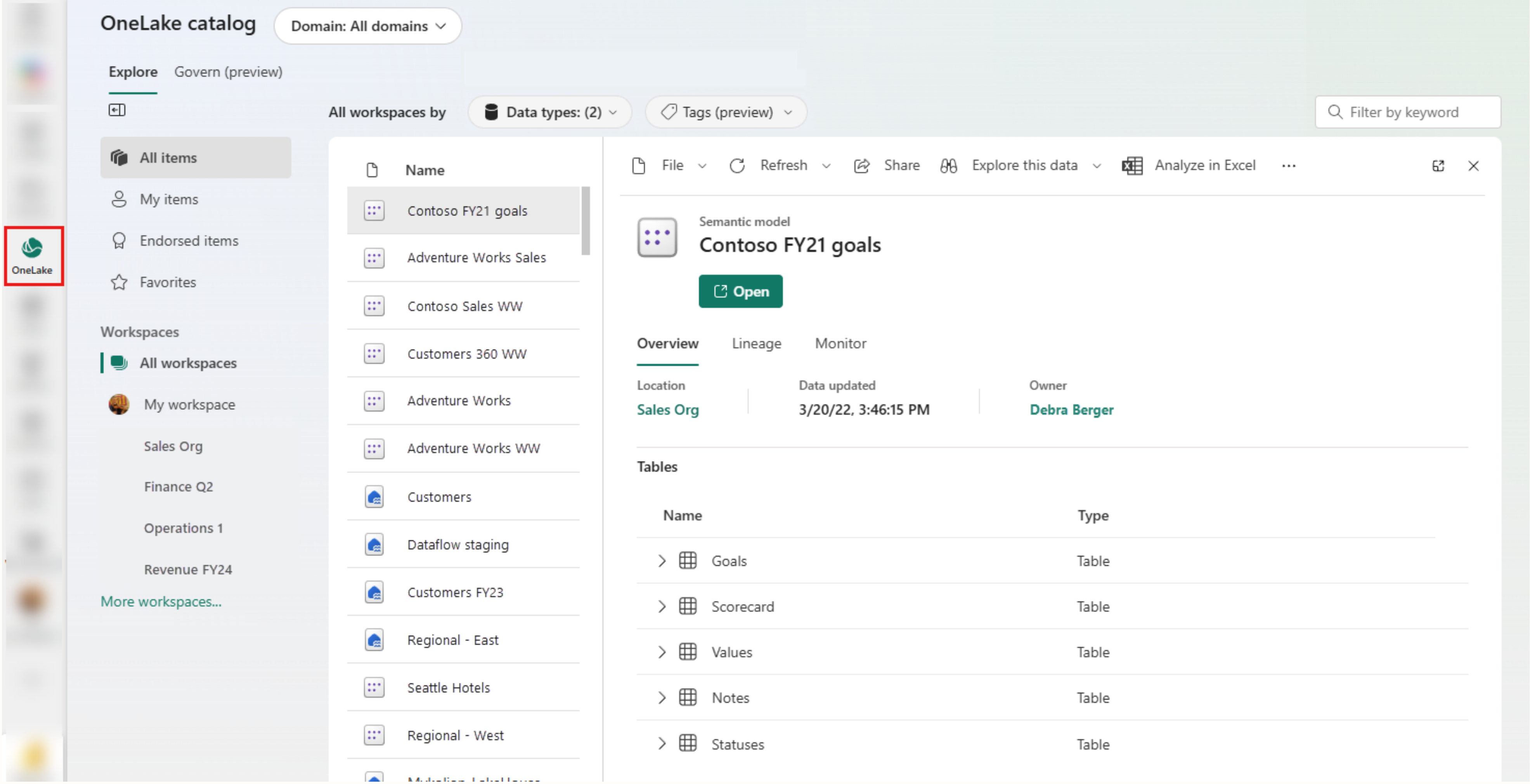


End-to-end security scenario – Medallion architecture



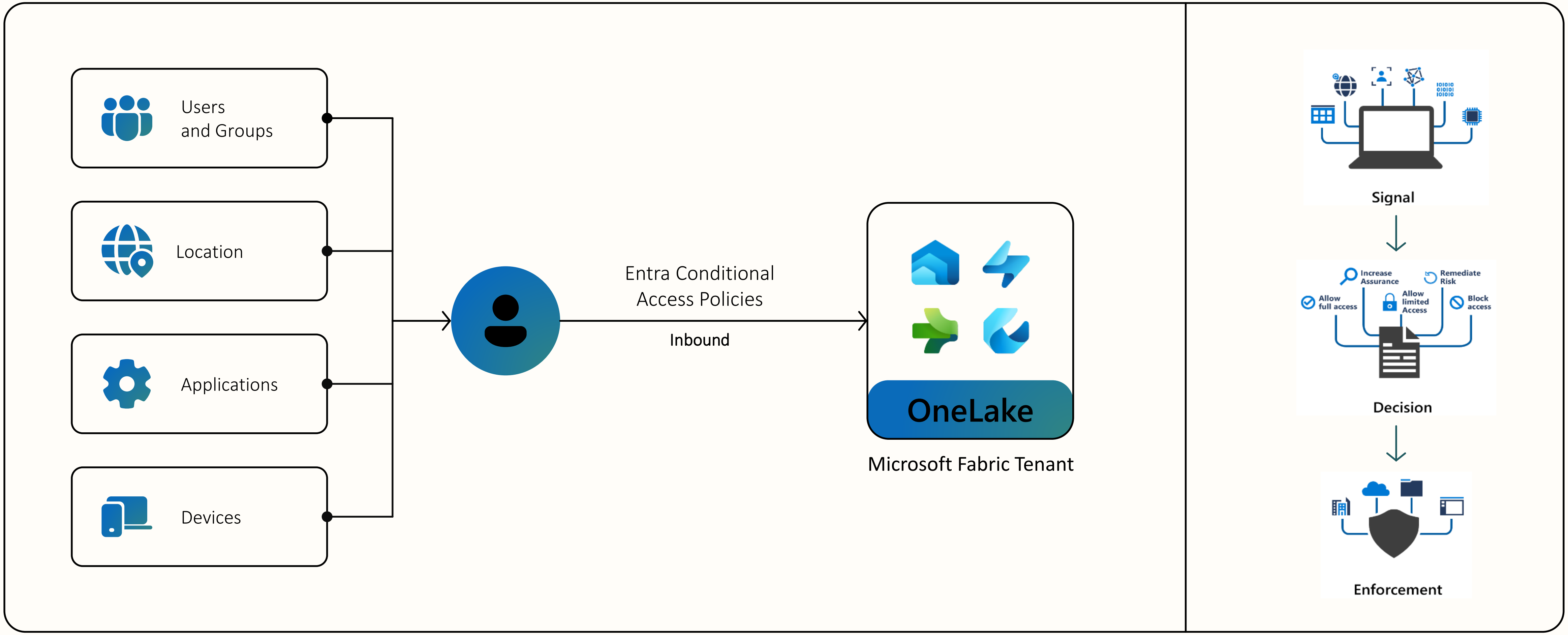
OneLake Catalog

Govern your data assets in Fabric



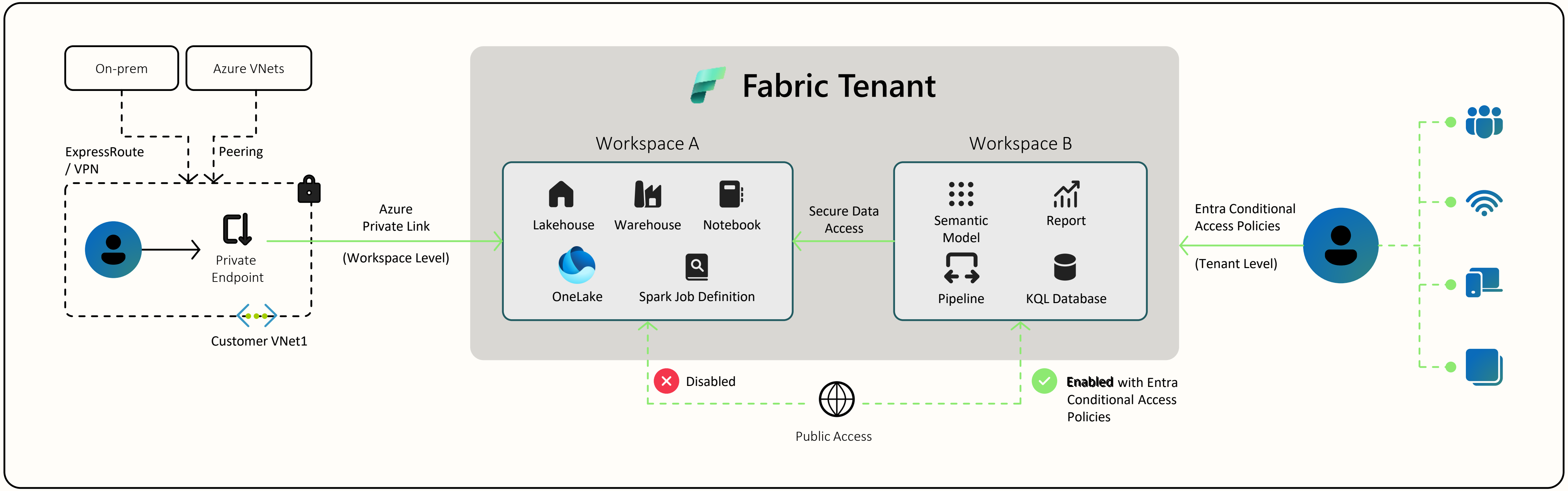
Conditional Access Policies

Decisions – Block, Grant, Require MFA



Workspace Private Link

Perimeter Network Security for your workspace



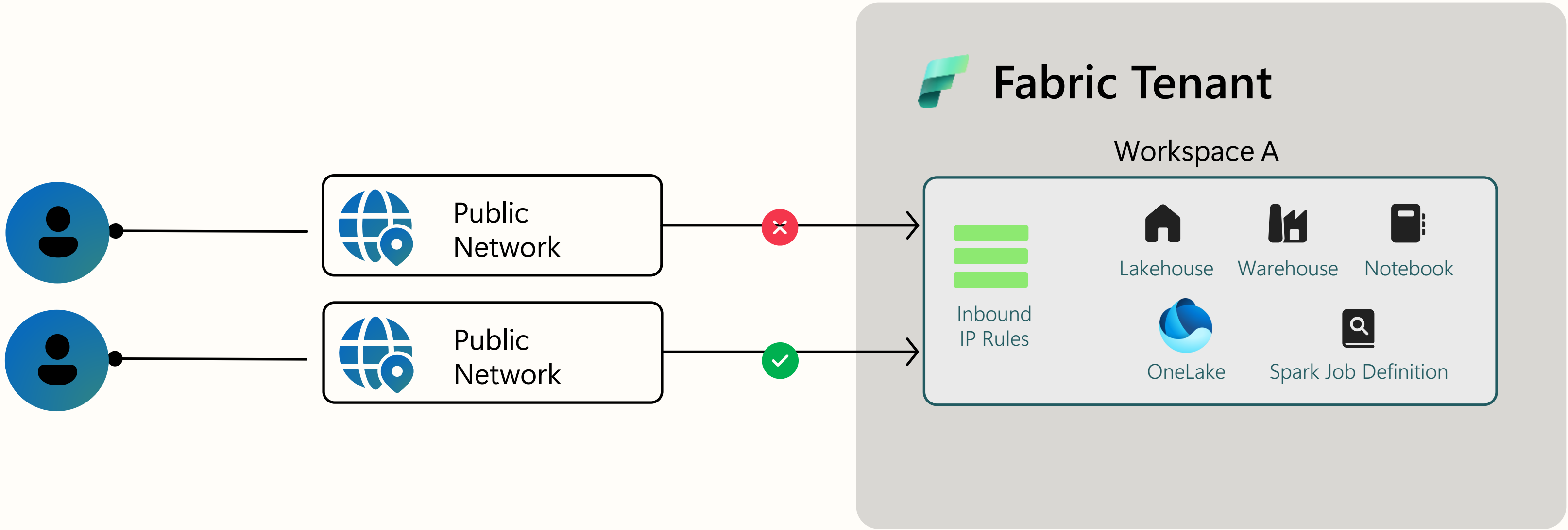
Selected workspaces can be protected using Private Links and closed from public internet.

Create a secure connection between public and private workspaces using private data access.

Public workspaces can be secured using Entra policies or IP filtering to use Power BI.

Workspace Private Link

Perimeter Network Security for your workspace



What it means:

1. Fabric workspace is accessible only from specific public IP or IP ranges
2. Simple setup

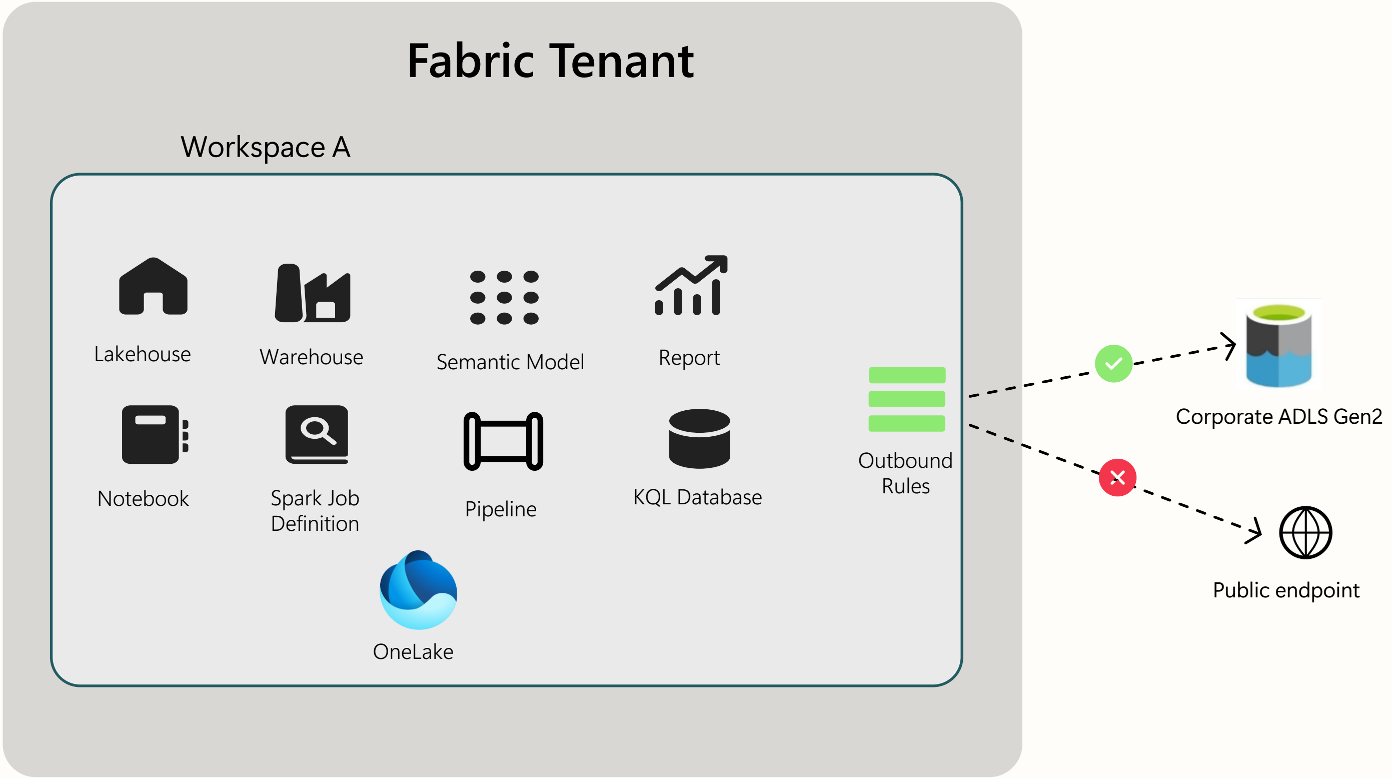
Outbound Access Protection

Define allowed destinations

Workspace level control to restrict outbound access to public network

Restrict outbound connections to permitted destinations only

Use in conjunction with other security features to achieve robust Data Exfiltration Protection

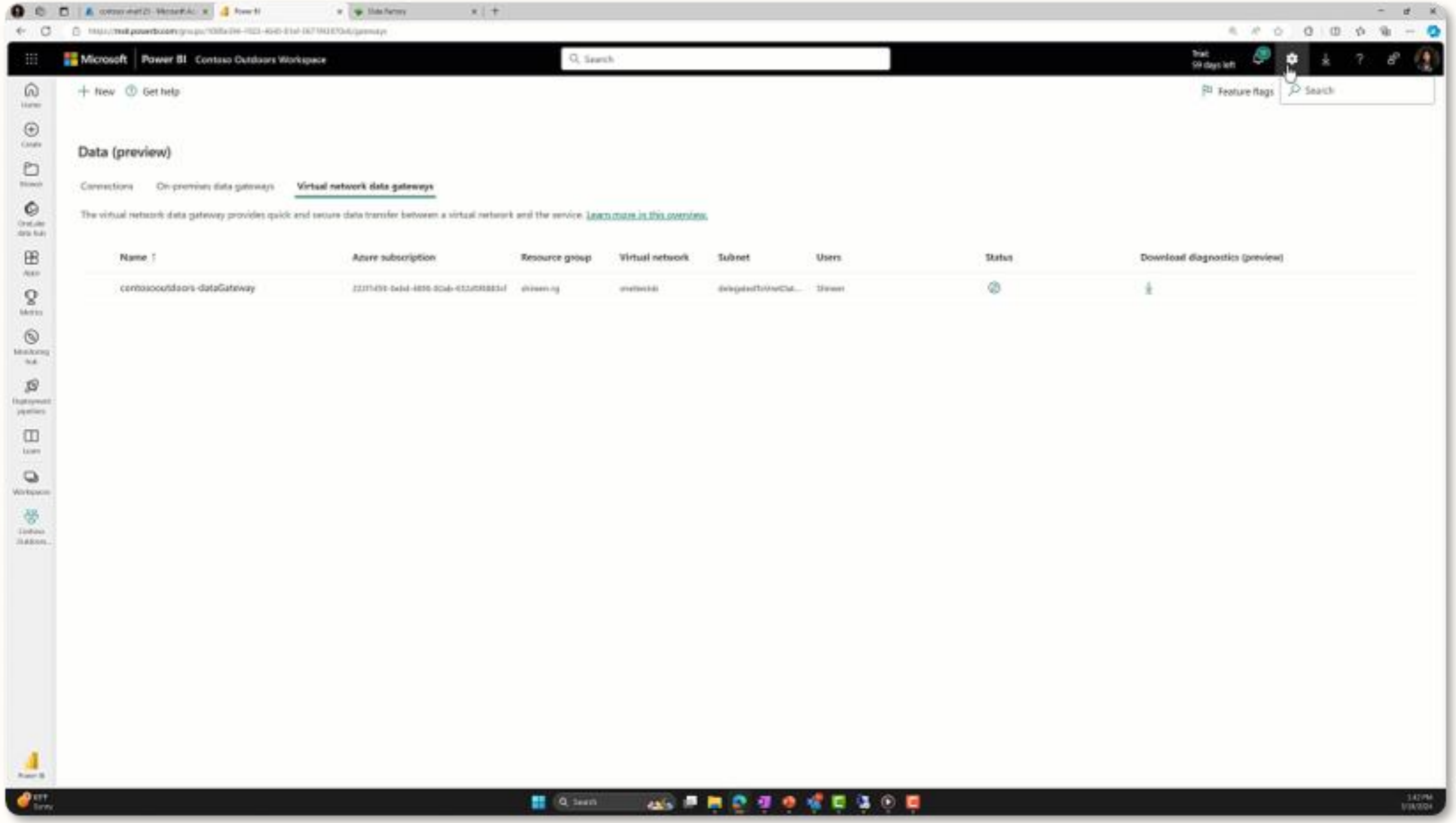


Vnet & On Prem Gateways

Secure connectivity to your Azure and private data services

VNet Gateways enable network security compliant access to secured resources

On Prem Gateways enable line of sight to on-prem data sources

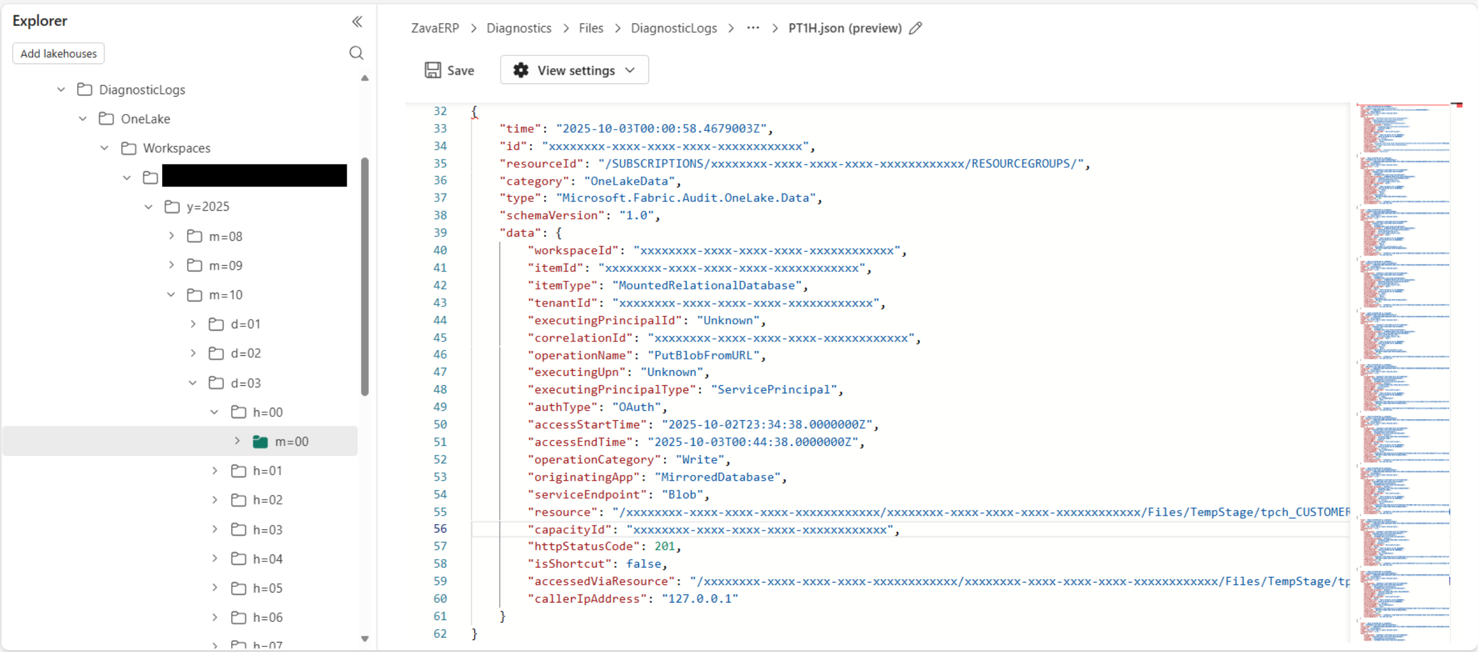


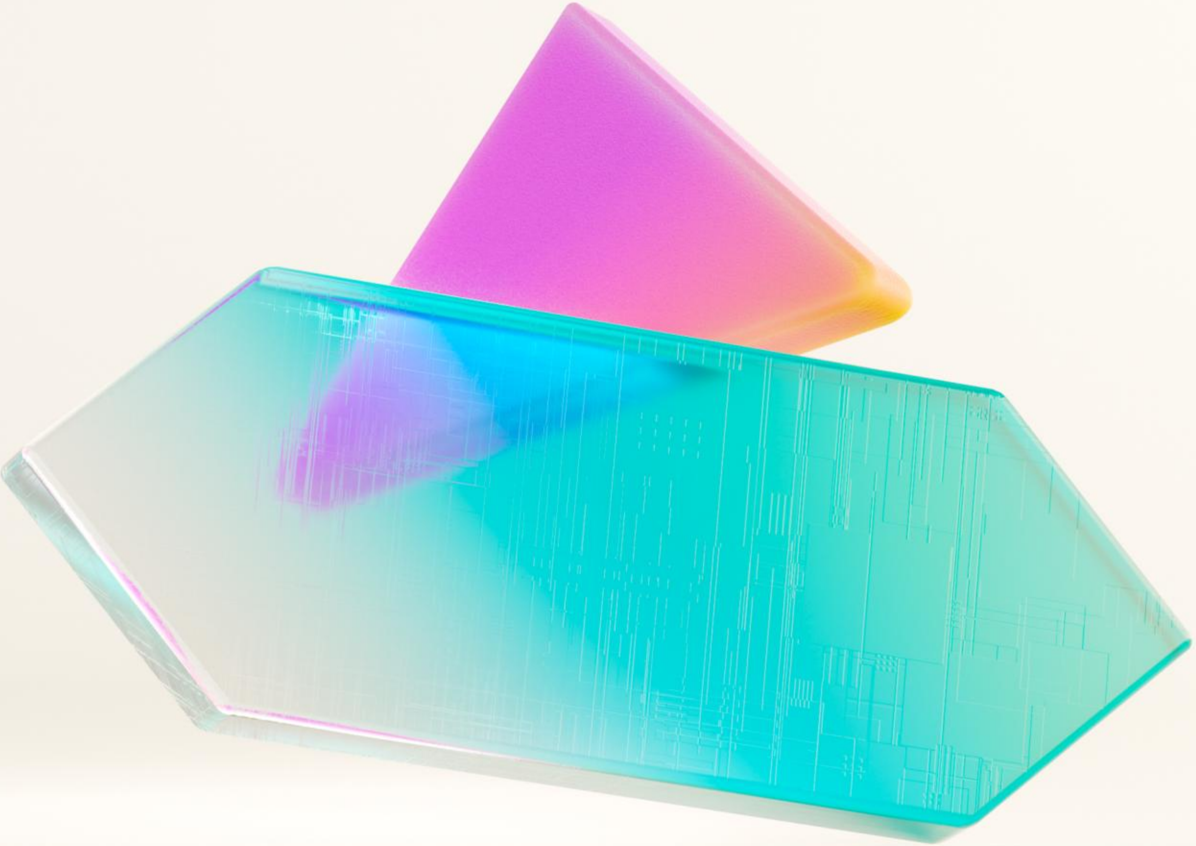
Onelake Diagnostics

Check what happened on your data

Enable this feature on workspace level and track actions on data in this workspace.

Combine with OneLake events to be directly informed and act based on events on any given scenario.





How to get started

Microsoft Fabric

Pricing options

Pay as you go

Only pay for the uptime of any given Fabric capacity. You don't need it? Just turn it off and start saving.

Reserved Capacity

As with other services in Azure you are able to commit to a capacity reservation for 1 year to receive discounts.

Mixed mode

Your data engineering is rather unpredictable? Make it pay as you go and have anything else pay against your reserved capacity.

… and how do I know what's best?

Start with pay as you go. Simply deploy a Fabric capacity and get started. Once you have your first projects running, lets check the usage patterns, optimize those and select appropriate strategy.

Microsoft Fabric simplicity

Microsoft Fabric is a unified product for all your data and analytics workloads. Rather than provisioning and managing separate compute for each workload, with Microsoft Fabric, your bill is determined by two variables: the amount of compute you provision and the amount of storage you use.

COMPUTE

A shared pool of capacity that powers all capabilities in Microsoft Fabric.

Pay-as-you-go and 1-year Reservation.

STORAGE

A single place to store all data.

Pay-as-you-go ($ per GB/month).

How it works

Capacity units (CUs) = Compute power

Capacity units (CUs) are units of measure that represent a pool of compute power needed. Compute power is required to run queries, jobs, or tasks.

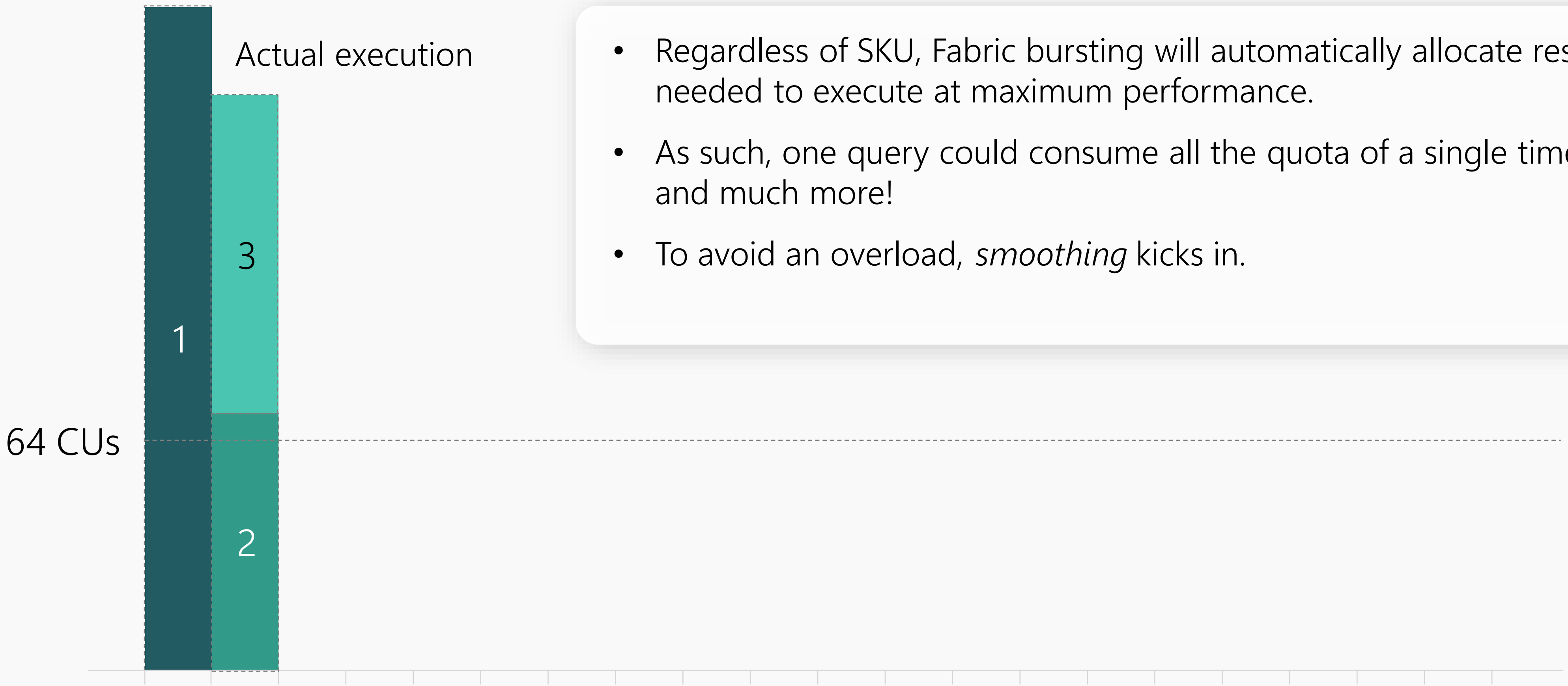
CU Consumption

The CU consumption is highly correlated to the underlying compute effort needed for the tasks performed by the capability during the processing time.

Each capability, such as Power BI, Spark, Data Warehouse, with the associated queries, jobs, or tasks has a unique consumption rate.

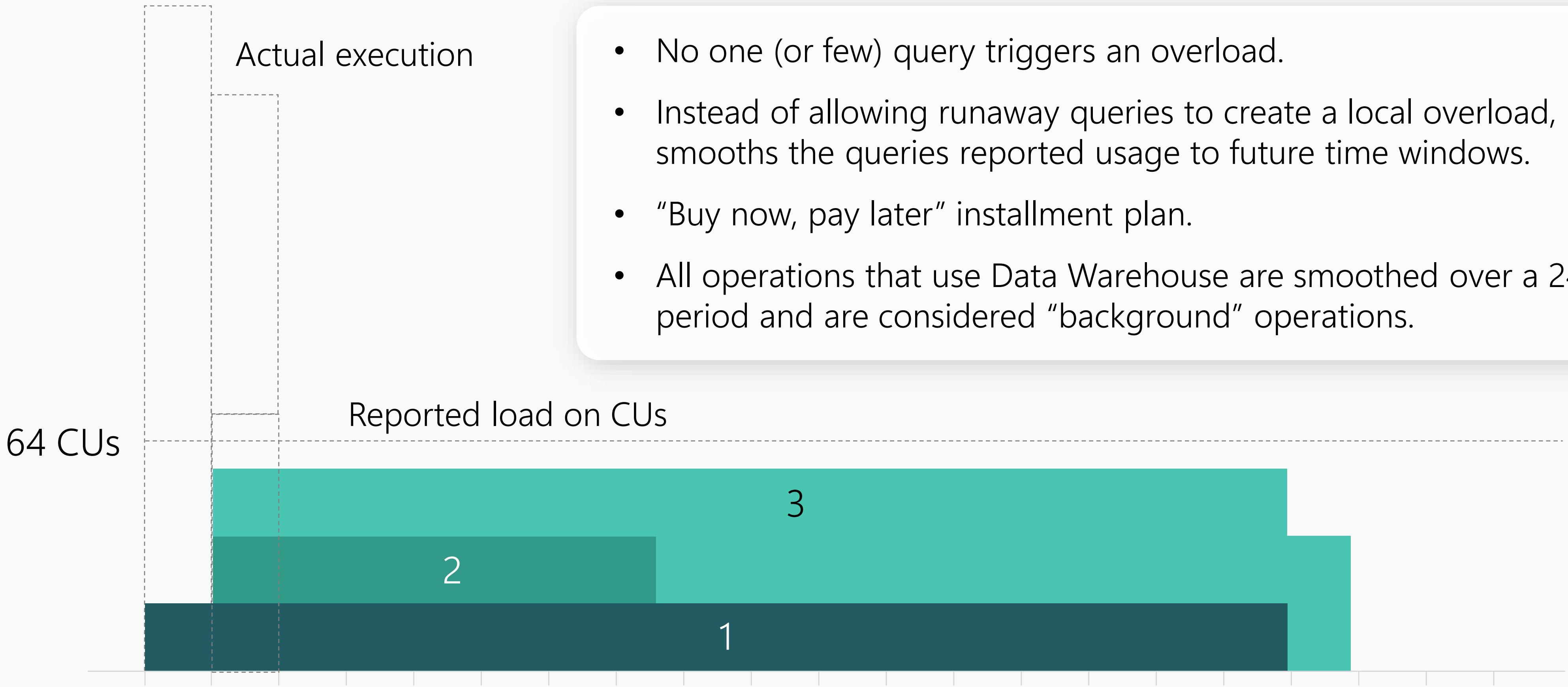
Note: For additional documentation on purchasing experience and capacity models, please see here: aka.ms/fabriclicensing

Bursting



- Regardless of SKU, Fabric bursting will automatically allocate resources as needed to execute at maximum performance.
- As such, one query could consume all the quota of a single time window and much more!
- To avoid an overload, *smoothing* kicks in.

Smoothing



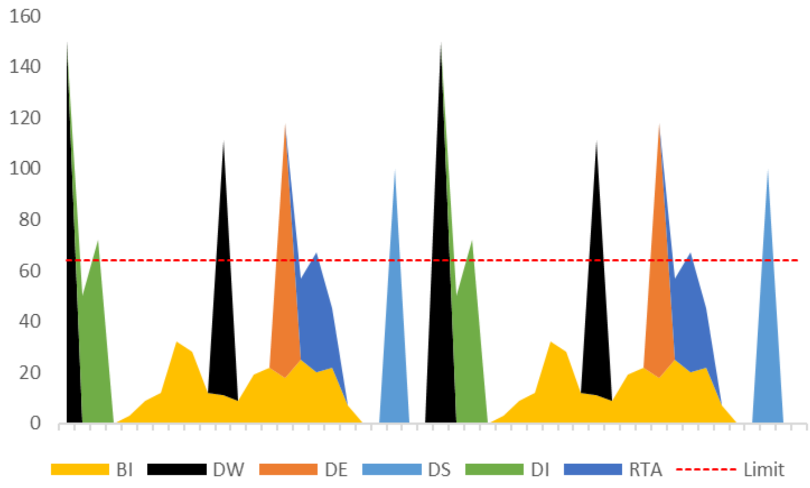
- No one (or few) query triggers an overload.
- Instead of allowing runaway queries to create a local overload, Fabric smooths the queries reported usage to future time windows.
- “Buy now, pay later” installment plan.
- All operations that use Data Warehouse are smoothed over a 24-hour period and are considered “background” operations.

Microsoft Fabric

Smoothing before and after

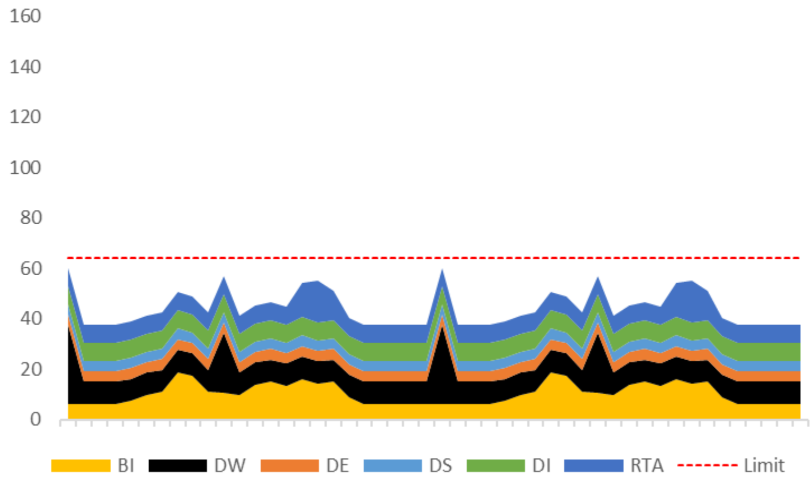
Before Smoothing

- Shows the **actual** load of multiple workloads before smoothing
- *Bursting* accelerates jobs execution by resource boosting
- The capacity is overloaded 25% of the time
- Some of the overloads are more than double the limit (*bursting*…)
- There are periods of no/low usage



After Smoothing

- Shows the **reported** load against the capacity limits
- There is no overload and consumption is more stable
- The smoothing of usage fills in gaps



Fabric capacity pricing

COMPUTE

Fabric capacity is priced uniquely across regions. The following table shows the pricing at **US West 2** for reference. Fabric capacity can be purchased at Azure portal. Visit Fabric pricing page for more details.

1 CU pay-as-you-go price at US West 2 *$0.18/hour*

SKU	Capacity unit (CU)	Pay-as-you-go (monthly)	Reservation (monthly) *~40.5% saving over Pay-as-you-go*
F 2	2	$262.8	$156.334
F 4	4	$525.6	$312.667
F 8	8	$1,051.2	$625.334
F 16	16	$2,102.4	$1,250.667
F 32	32	$4,204.8	$2,501.334
F 64	64	$8,409.6	$5,002.667
F 128	128	$16,819.2	$10,005.334
F 256	256	$33,638.4	$20,010.667
F 512	512	$67,276.8	$40,021.334
F 1024	1024	$134,553.6	$80,042.667
F 2048	2048	$269,107.2	$160,085.334

Note:

1. 1 CU PAYGO monthly rate calculation: $0.18*730 =$131.4. F2 =$131.4*2=$262.8
2. 1 CU RI monthly rate calculation: Round ($0.18* (1-0.405)*730*12,0)/12=~$78.166...F2 RI = ~$78.166...*2=~$156.334
3. Power BI Pro license is required for all Power BI Premium ("P") and Fabric Capacity ("F") SKUs to publish Power BI content to Microsoft Fabric. Enabling content consumers to review and interact with Power BI reports without additional paid per-user licenses is available at P1 and above (and F64 and above).

OneLake pricing

OneLake is a data lake built into Microsoft Fabric and provides a single place to store all organizational data. Data storage is charged at a rate of $ per GB per month and priced uniquely across regions.

Data Storage

Type	Pay-as-you-go price at West US 2
OneLake storage	$0.023 per GB/month
OneLake BCDR storage	$0.0414 per GB/month
OneLake cache	$0.2 per GB/month

Data Transfer & Internet Egress

Cross-region data transfer network charges may apply based on source/destination of each storage access. Learn more at the Bandwidth Pricing.

Note: OneLake cache is billed for KQL cache storage and Activator data retained.

Translation for existing Power BI Premium per capacity Customers

COMPUTE

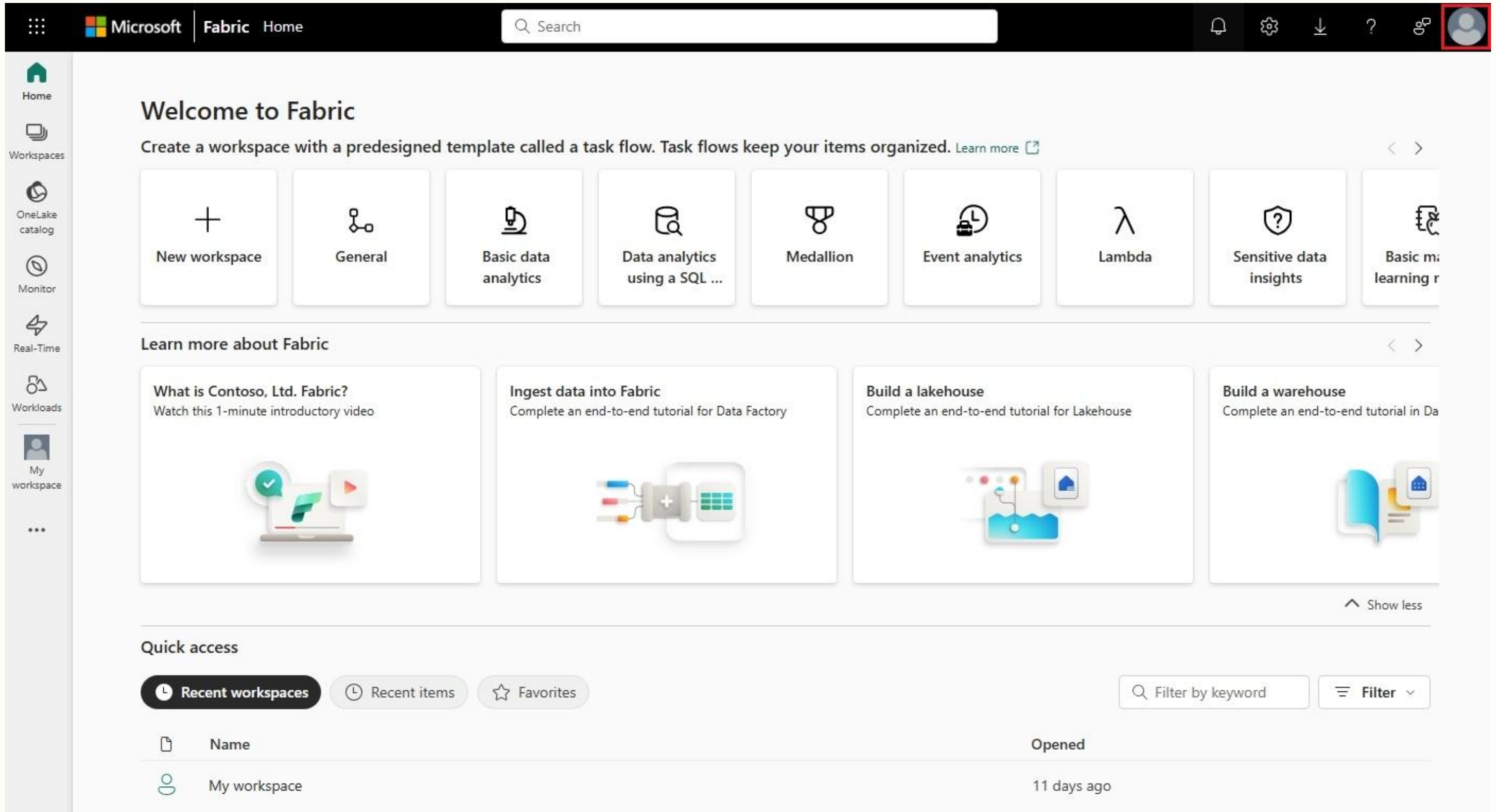
Today	
SKU	**PBI vCore**
P1	8
P2	16
P3	32
P4	64
P5	128

Microsoft Fabric Generally Available	
SKU	**CU**
F 64	64
F 128	128
F 256	256
F 512	512
F 1024	1024

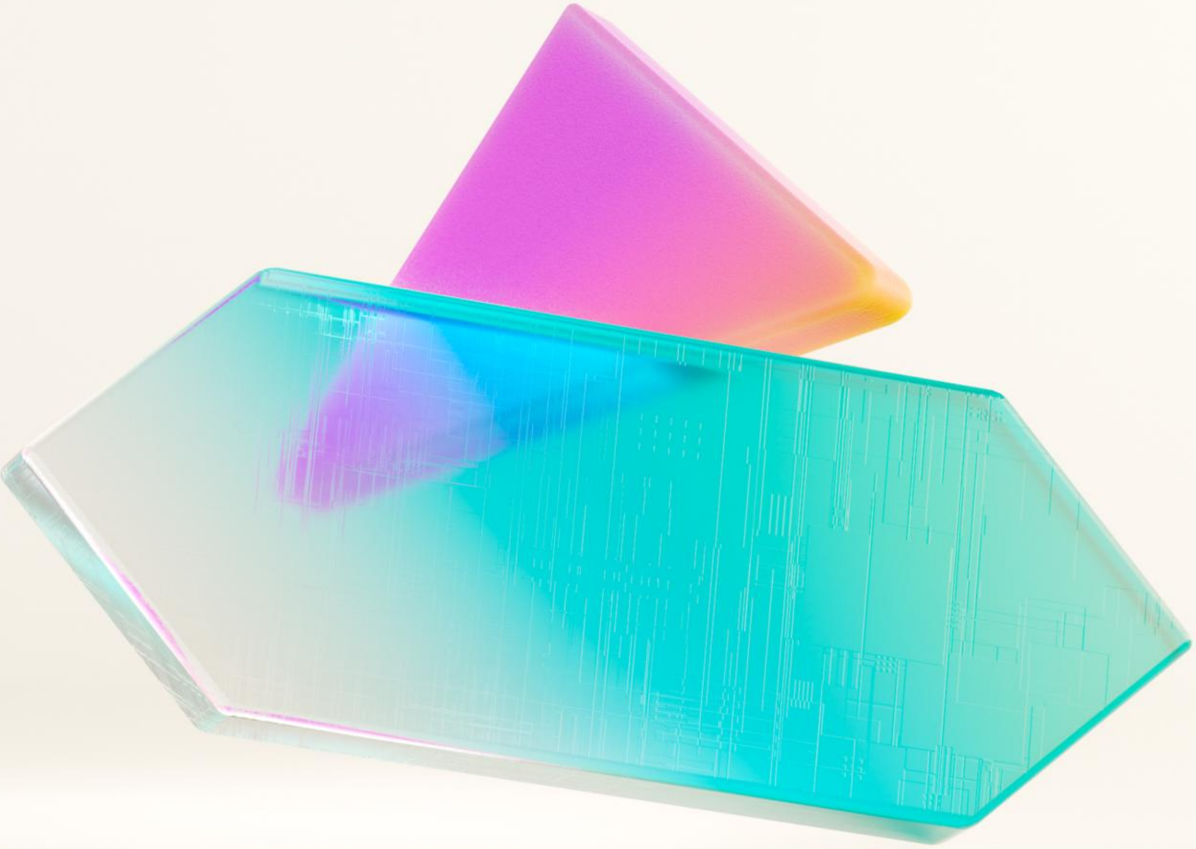
- Power BI Premium P SKUs get **access to the full Fabric capabilities**
- 8 CUs are providing the same compute power of 1 Power BI Premium v-core for Power BI workloads. That is, **F64 provides the equivalent compute power of Power BI Premium P1**

For additional documentation on Fabric licensing types, please see here: [Microsoft Fabric concepts – Microsoft Fabric | Microsoft Learn]

Fabric Trial



- Get your 60 days Fabric trial capacity within your tenant today.
- Test workloads, build first small MVPs, plan your architecture not on a drawing board but on the live object.



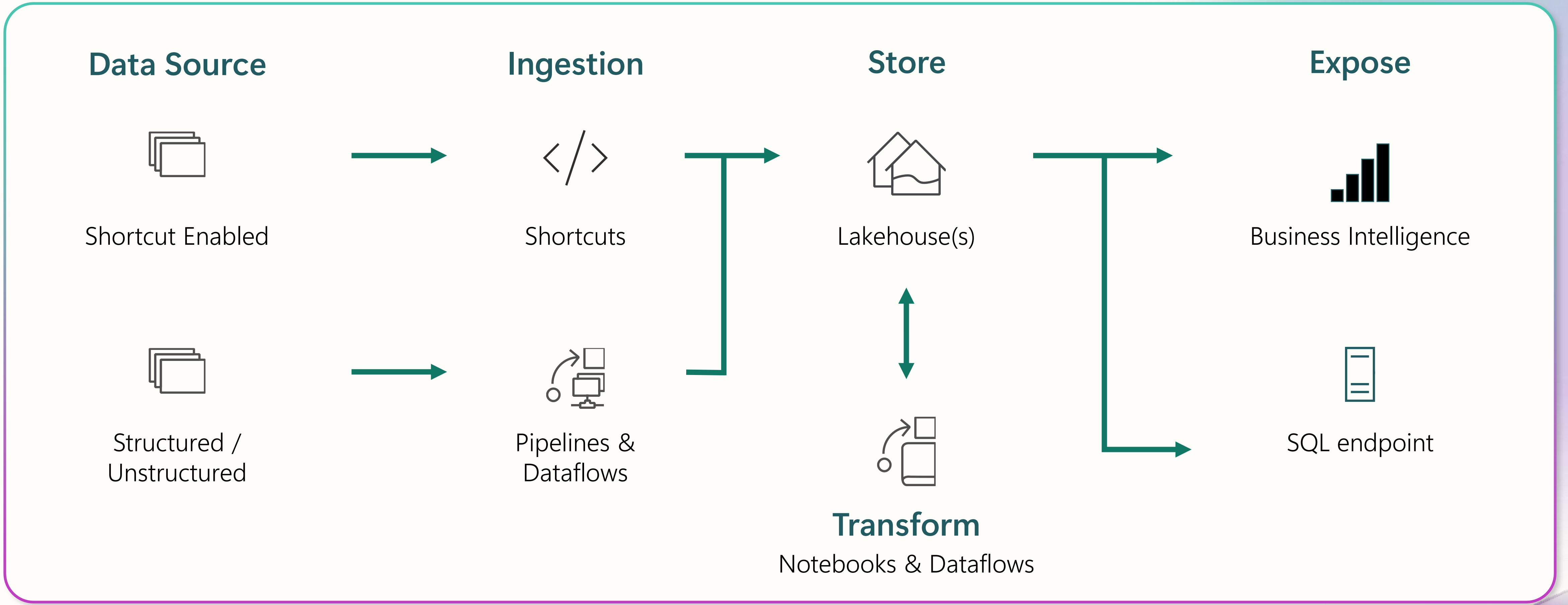
Data Engineering

Simplify complex operations with a modern data movement strategy

Today's data movement	Tomorrow's data movement
Replication-heavy Data copied multiple times before use	**Direct access** Eliminate redundant copies with data-on-demand
Delayed processing Insights slowed by nightly or scheduled batch loads	**Continuous pipelines** Stream data in real-time for faster decision making
Fragmented governance Policies and controls inconsistent across sources	**Unified governance** Consistent policies and security across all data sources
Complex operations High overhead moving data with slow time-to-value	**Automated orchestration** Streamlined management with faster time-to-insight

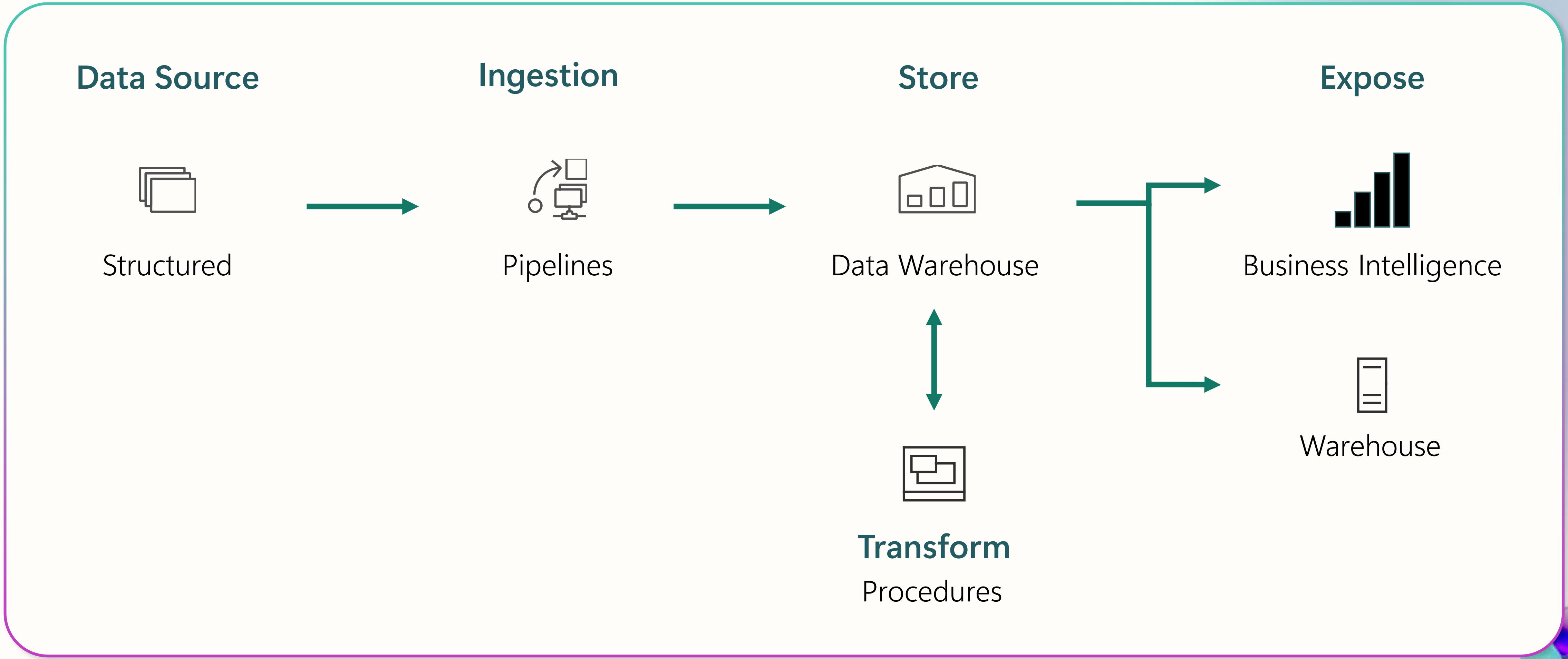
Lakehouse scenario

End-to-end analytics scenario



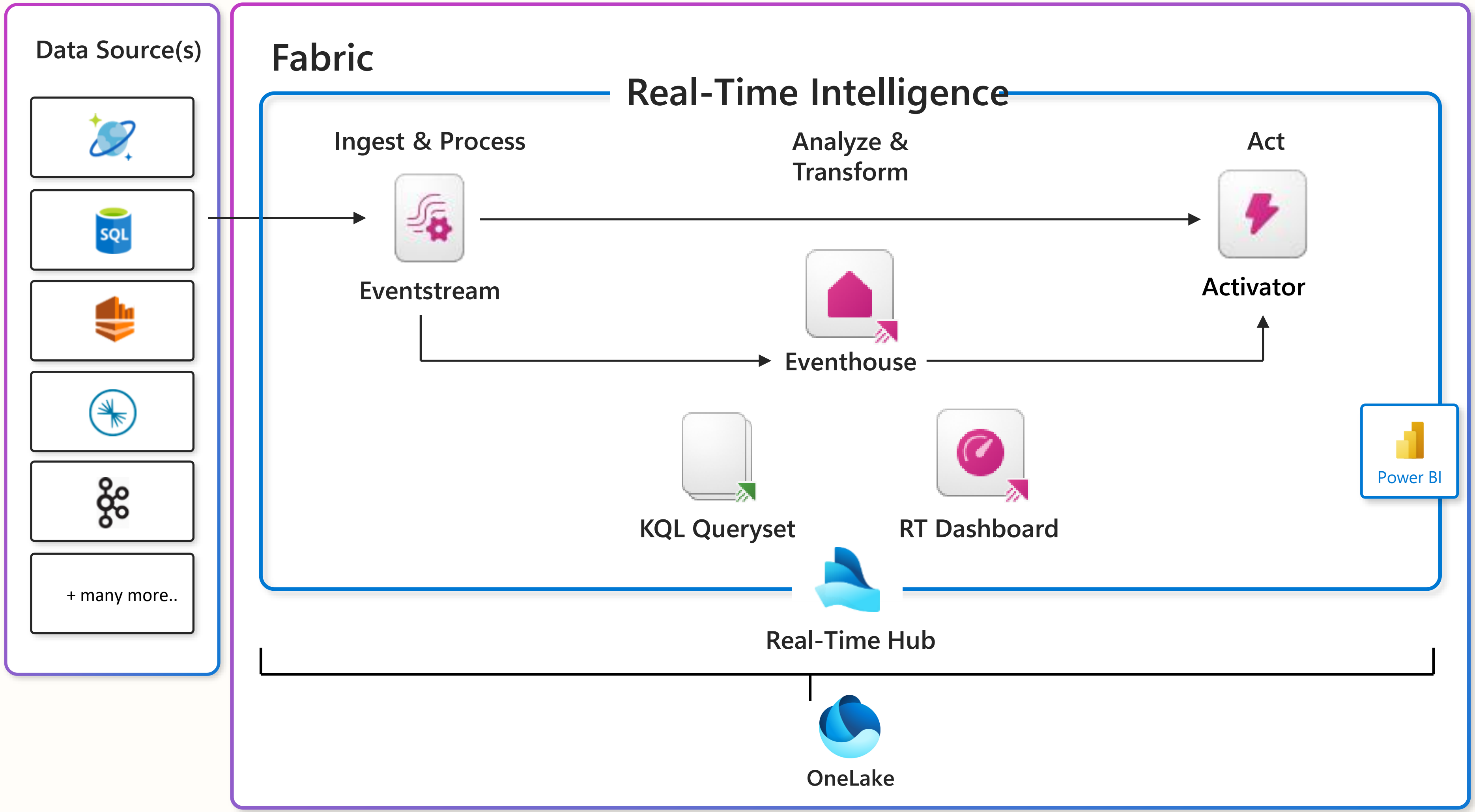
Data warehouse scenario

End-to-end analytics scenario



Real-Time Intelligence scenario

End-to-end analytics scenario



A zero copy, zero ETL approach hinges on flexible data ingestion

Mirroring

Gain near real-time data freshness by continuously syncing operational databases

Make data instantly analytics-ready in OneLake and eliminate complex ETL

Open mirroring

Extend flexibility with APIs and partner integrations to avoid custom pipelines

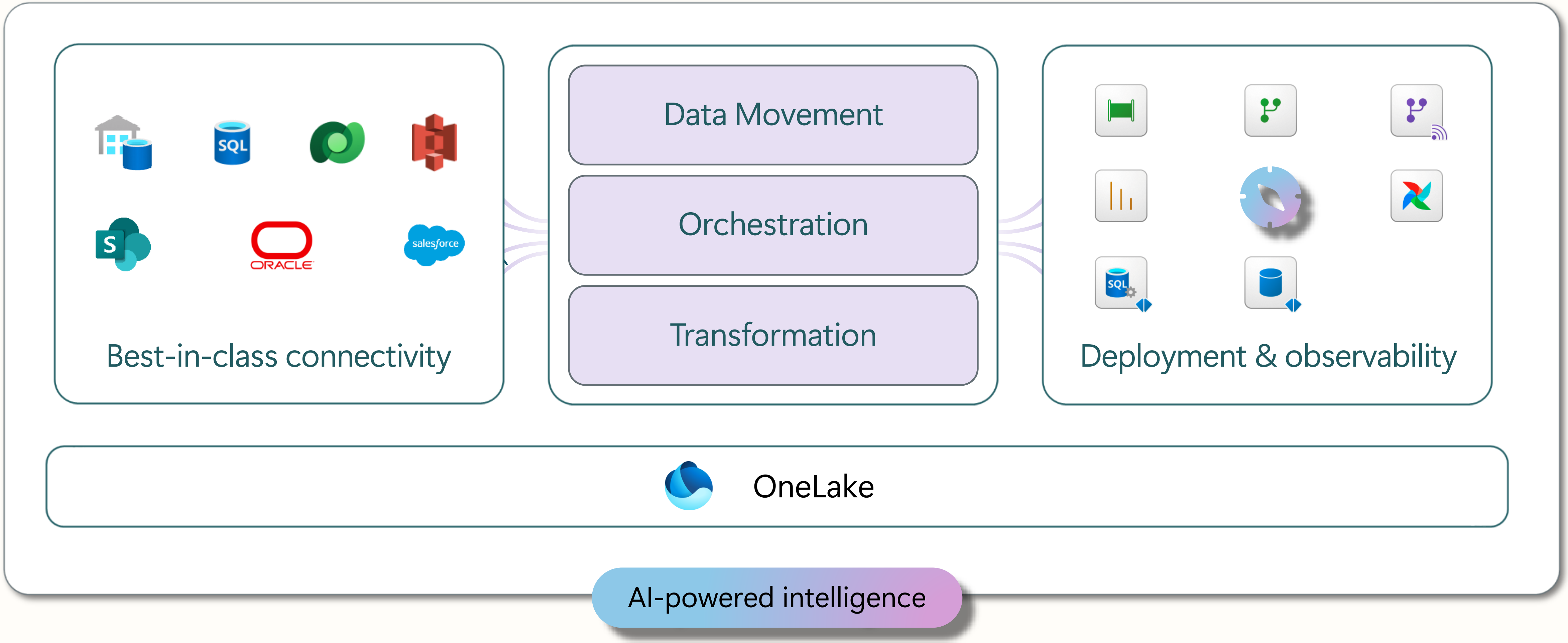
Simplify movement and ensure consistency by standardizing on open Delta format

Shortcuts

Access data across clouds and domains without duplication and instantly make it useable in OneLake

Reduce storage needs and latency by eliminating physical copies

Data Factory

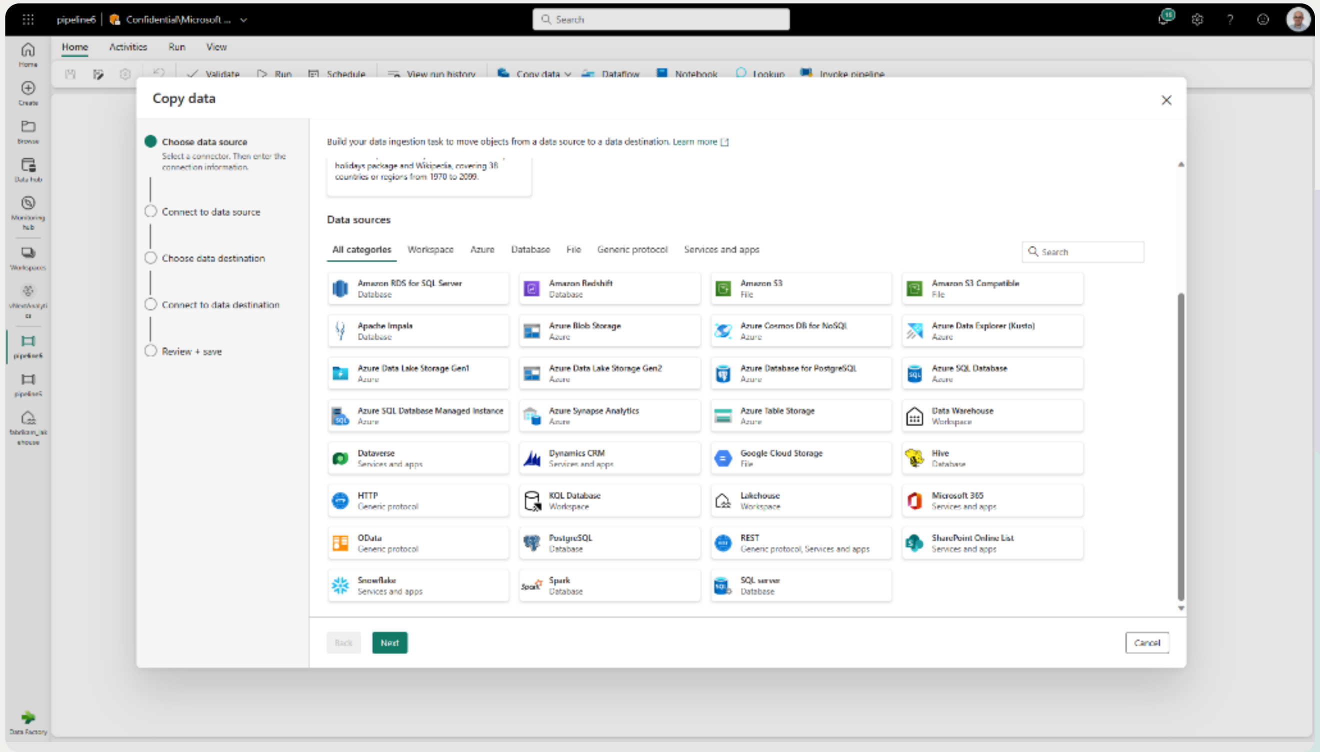


Connectors

New Connectors provide a low-code interface for ingesting data from a variety of data sources.

Connectors:

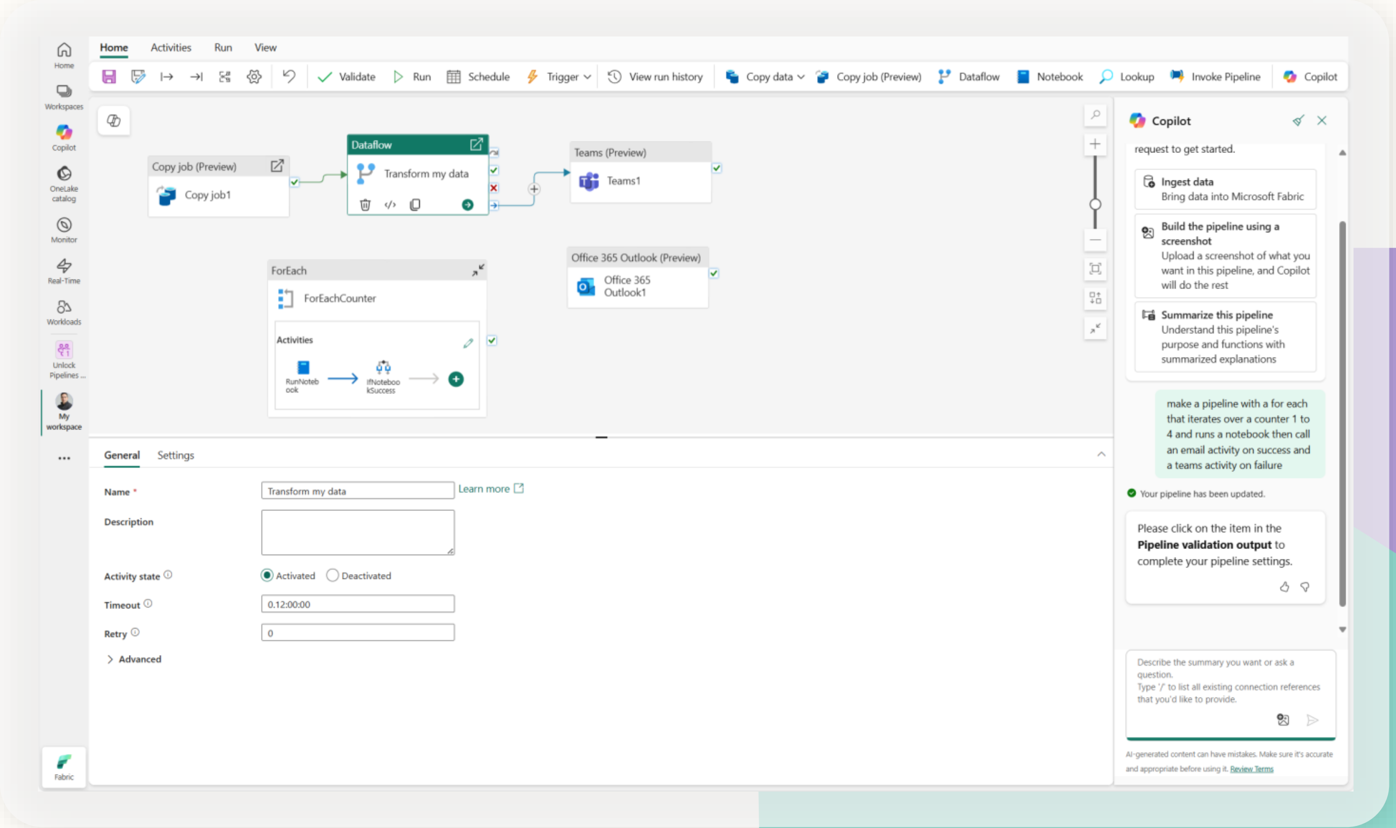
- Fabric native Connectors
- Connect to Azure and non-Azure resources
- Integrated with modern get data experience
- 170+ connectors across Data Factory
- Access to on-premises data
- Access protected data inside of a VNET



Data Pipelines

Data Pipelines enable powerful workflow capabilities at cloud-scale like building complex workflows, moving PB-size data, and defining sophisticated control flow pipelines.

Data pipelines can be used to build complex ETL and data factory workflows that can perform multiple different tasks at scale. Additionally, control flow capabilities are built into pipelines so you can build workflow logic which provide loops and conditional execution.



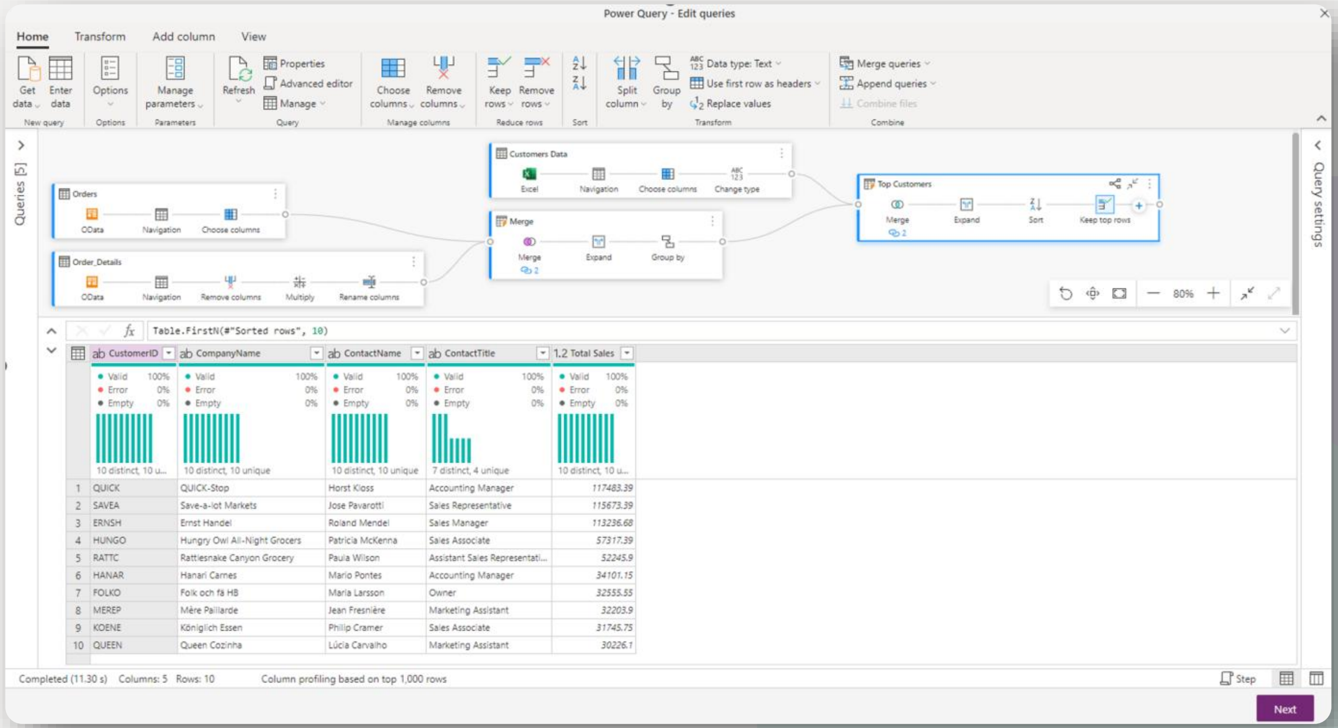
Dataflow

Dataflow provides a low-code interface for ingesting & transforming data from hundreds of data sources.

Dataflows quickly and easily unify disparate data sources, establish a more collaborative analytics approach, and promote more informed, agile decision making.

Key Capabilities:

- Accelerate data transformation with code-free dataflows
- Achieve high-scale ingestion & transformation using Fabric compute and Data Factory fast copy
- Load results of data transformations into multiple destinations (Azure SQL Databases, Lakehouse, SharePoint, etc.)



Notebook | Overview

Immersive authoring experience for data developers

Rich notebook capabilities including native Lakehouse integration, real-time collaboration with commenting support, auto-save support, lightweight scheduling and pipeline integration

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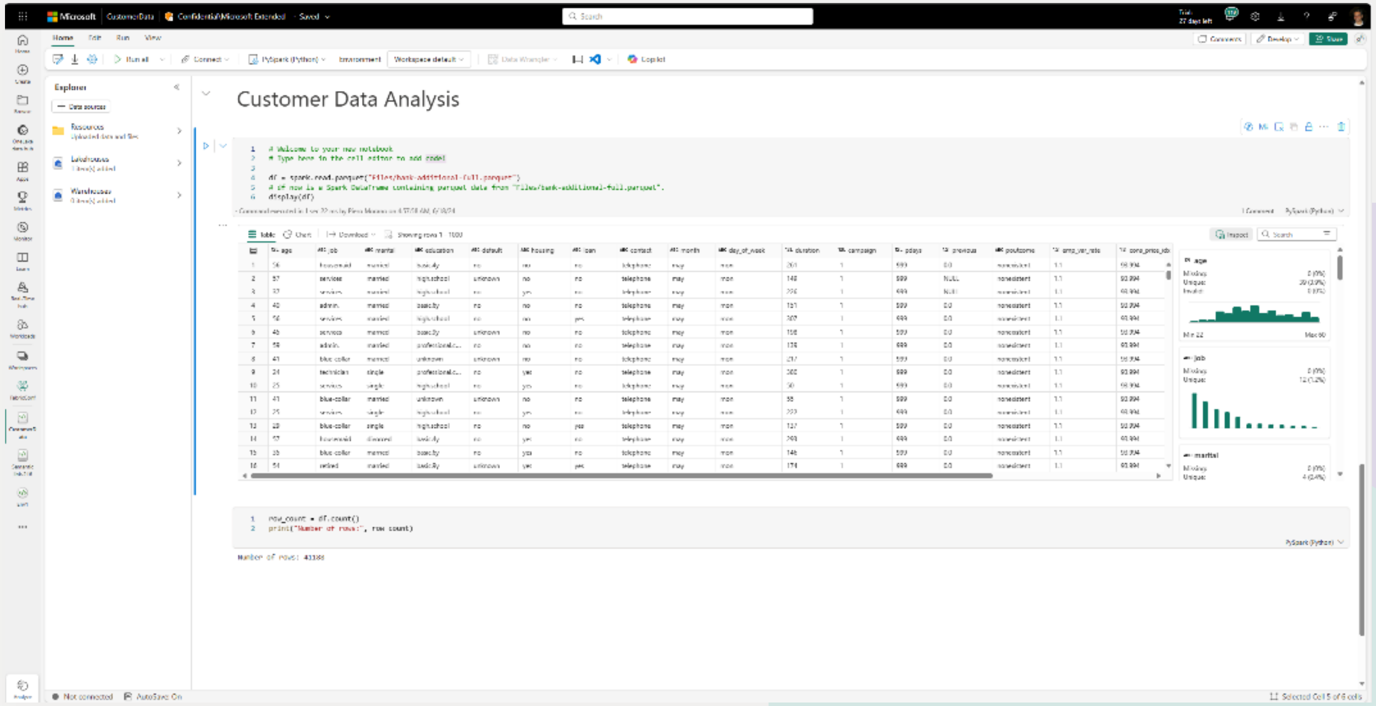
Key Capabilities:

- Modern data frame display, rich summary statistics & display, rich summary statistics & visualizations
- Advanced notebook development support with ability to reference notebooks in notebooks, and snapshots for easy troubleshooting
- In context monitoring complete with real time advice and error analysis
- Leverage Data Wrangler for low code data prep for Spark



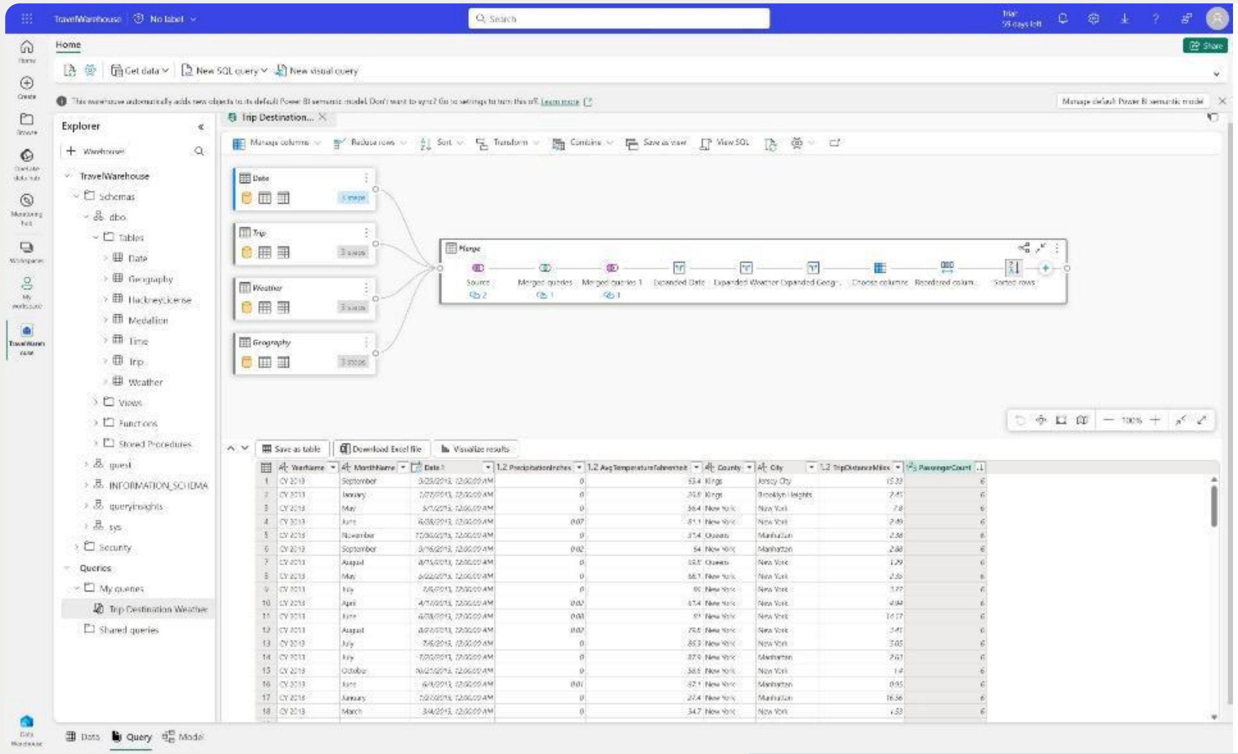
Virtual warehouses and cross database querying

Eliminate the need to duplicate data from any Lakehouse to any Warehouse

Customers can query data in the lake without the need for data movement, reducing costs

Key Capabilities:

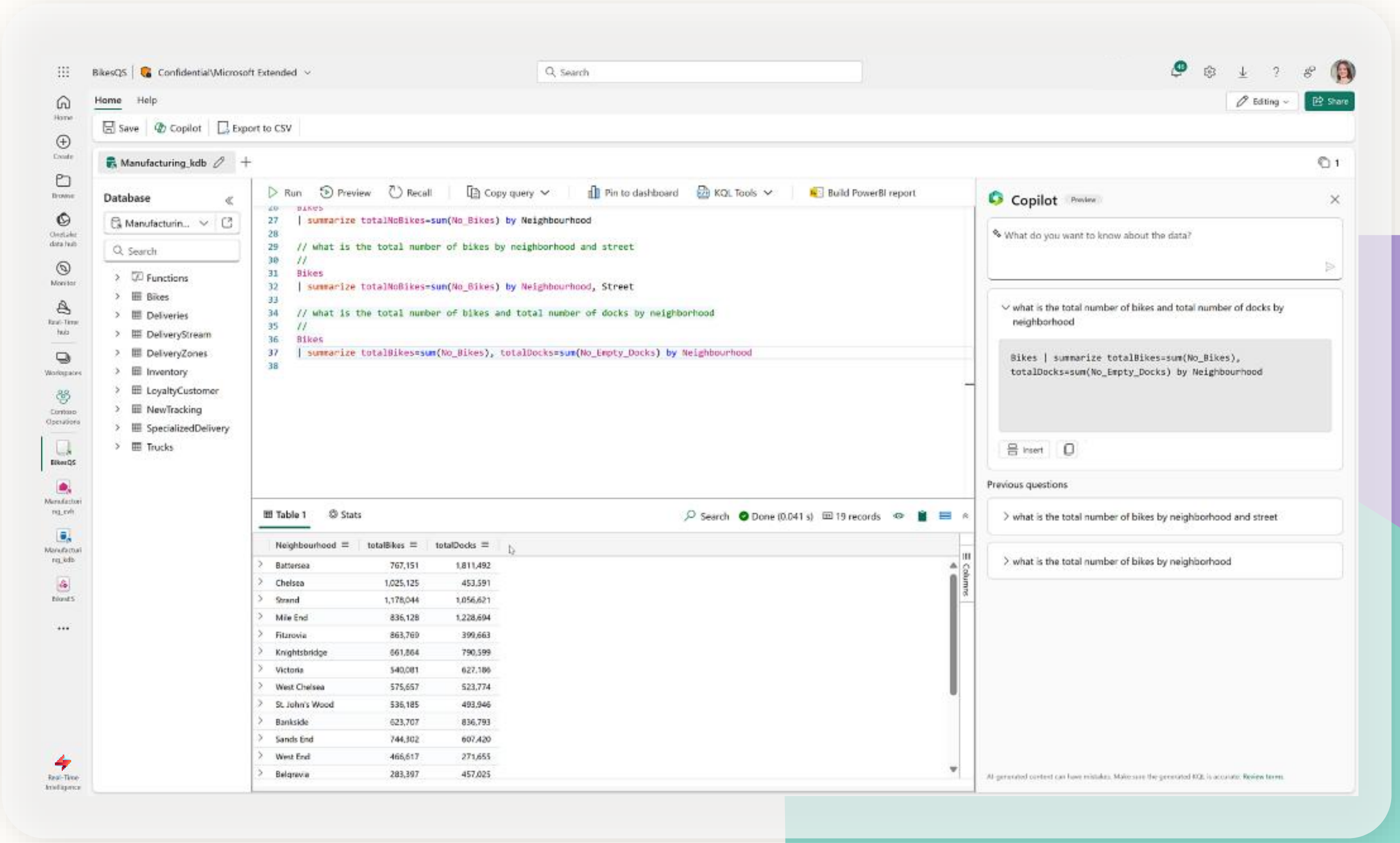
- Build virtual warehouses by creating Lakehouses with shortcuts to data in the Lake
- Query across Warehouse and Lakehouse SQL Endpoint with zero data movement
- Build queries visually using the Visual Query Editor



Real-time AI insights – and more to come!

Key Capabilities:

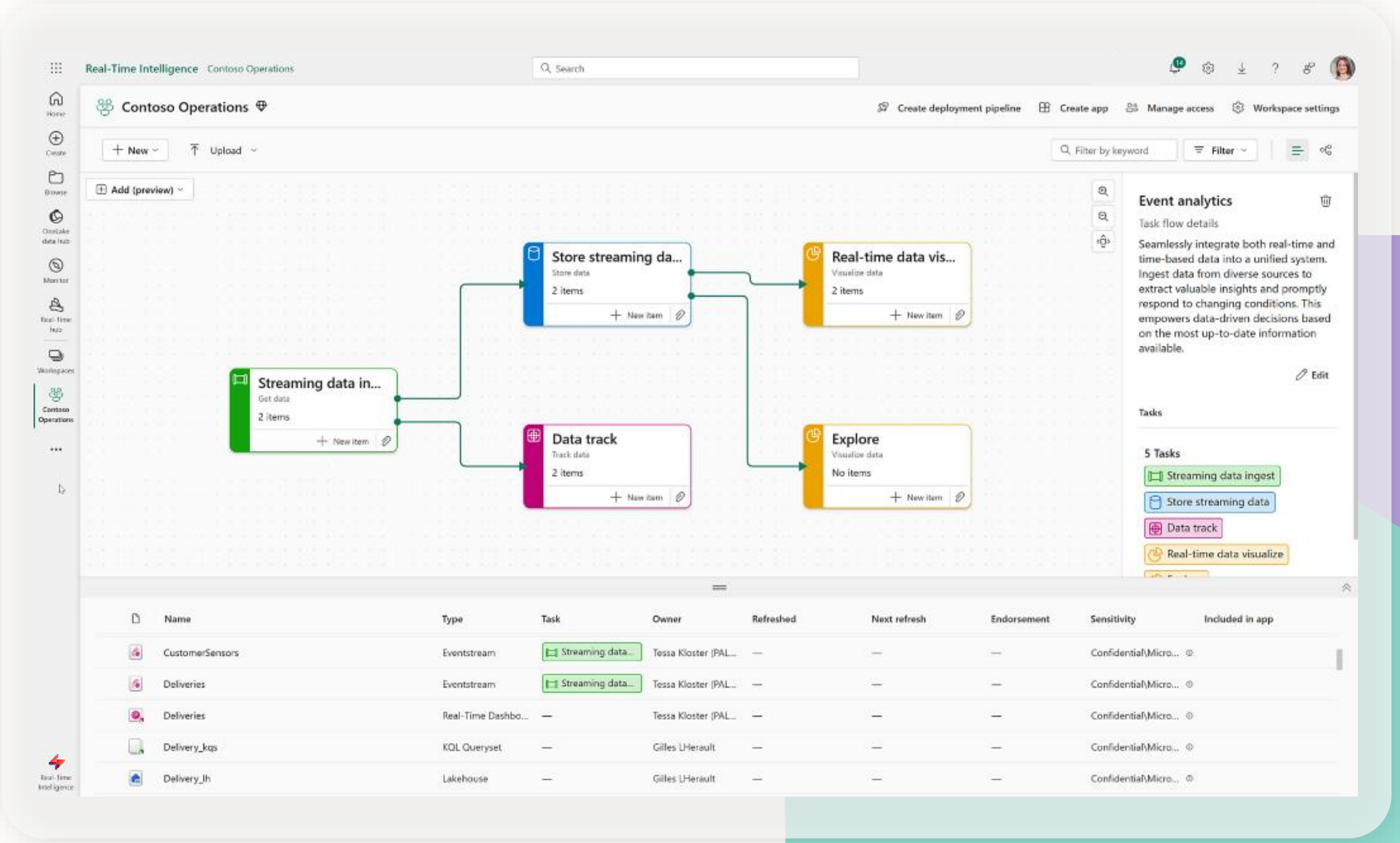
- Leverage Copilot to automate routine tasks and act as an interactive aide
- Easily generate KQL queries with copilot experience
- Monitor data and automatically drive alerts when anomalies are detected
- Use real-time insights to build and scale advanced ML models in Azure Machine Learning
- Create generative AI experiences on top of your time-oriented data with Azure AI Foundry

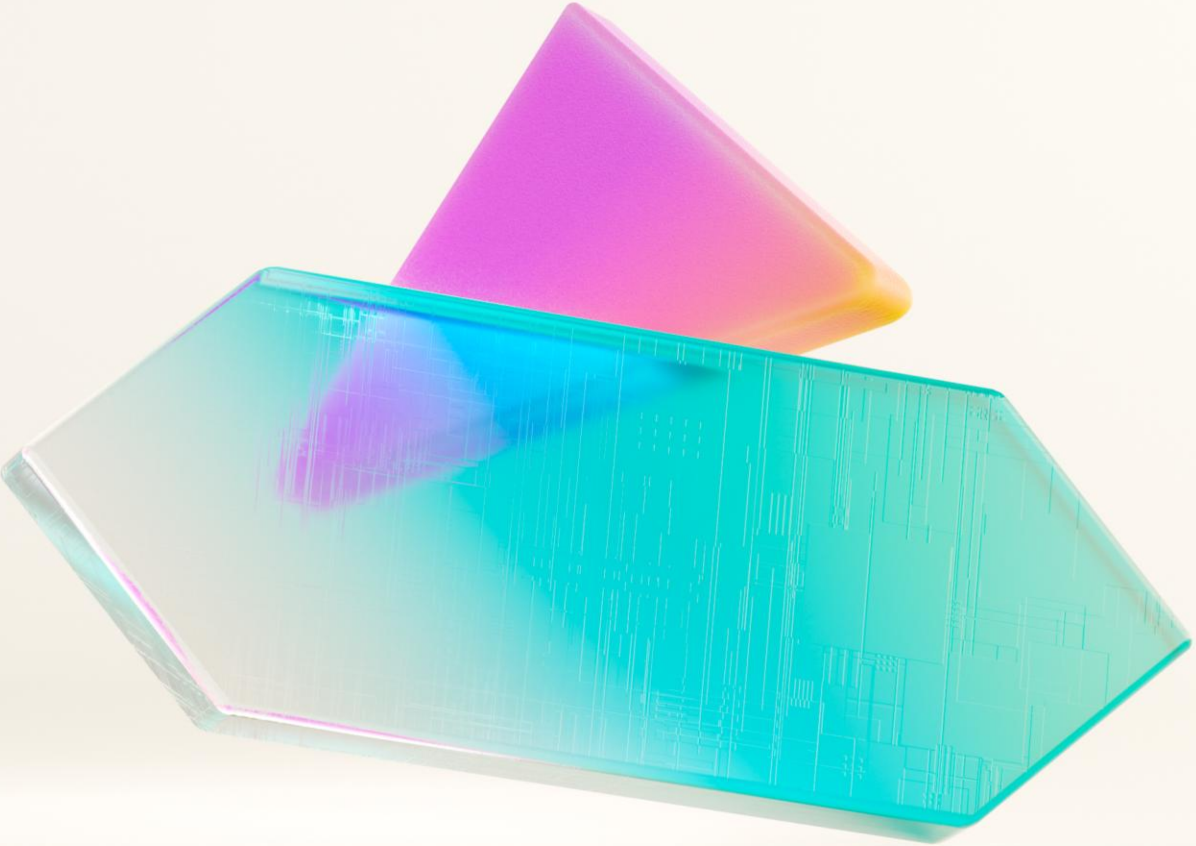


Work across full event life cycle

Key Capabilities:

- Gain deeper insight into streams with data preview and usage history
- Create triggers through simple, embedded experiences
- Open event streams to process and route events without writing any code
- Land in Eventhouse for further analysis
- Analyze and build lightweight models
- Create derived and enriched streams

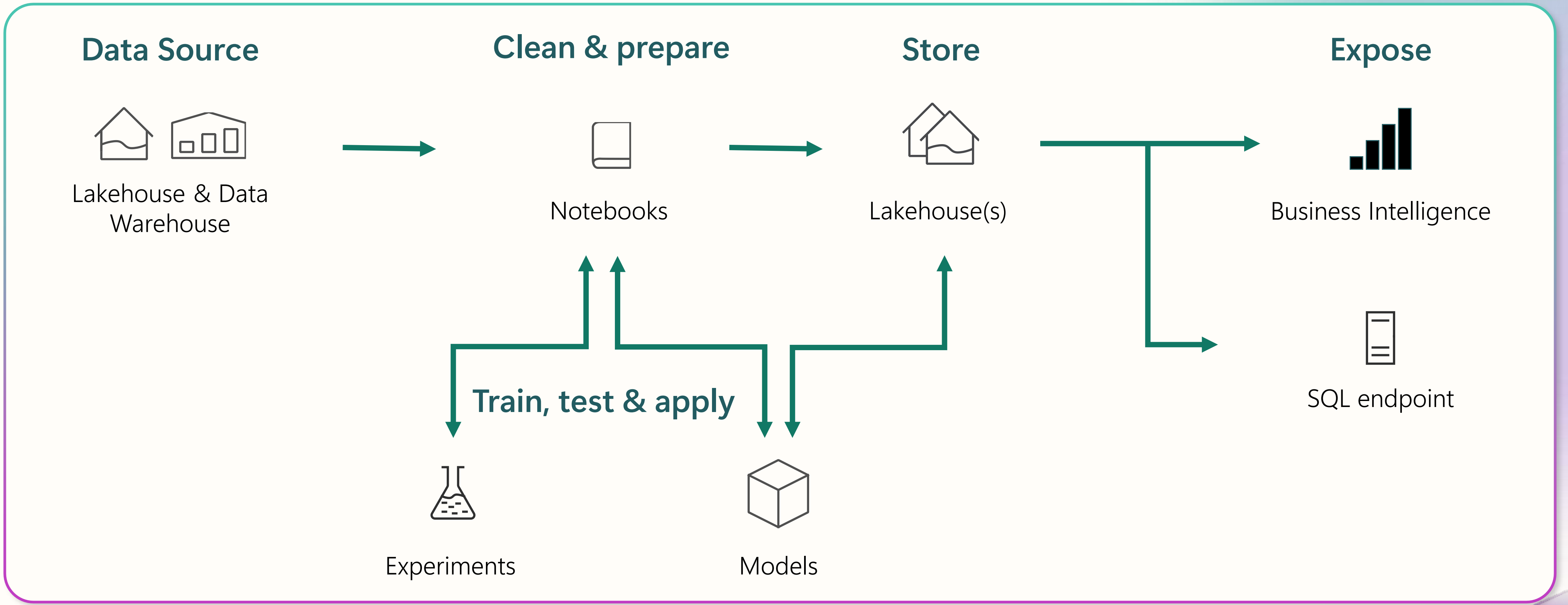




Data Science

Data science scenario

End-to-end analytics scenario

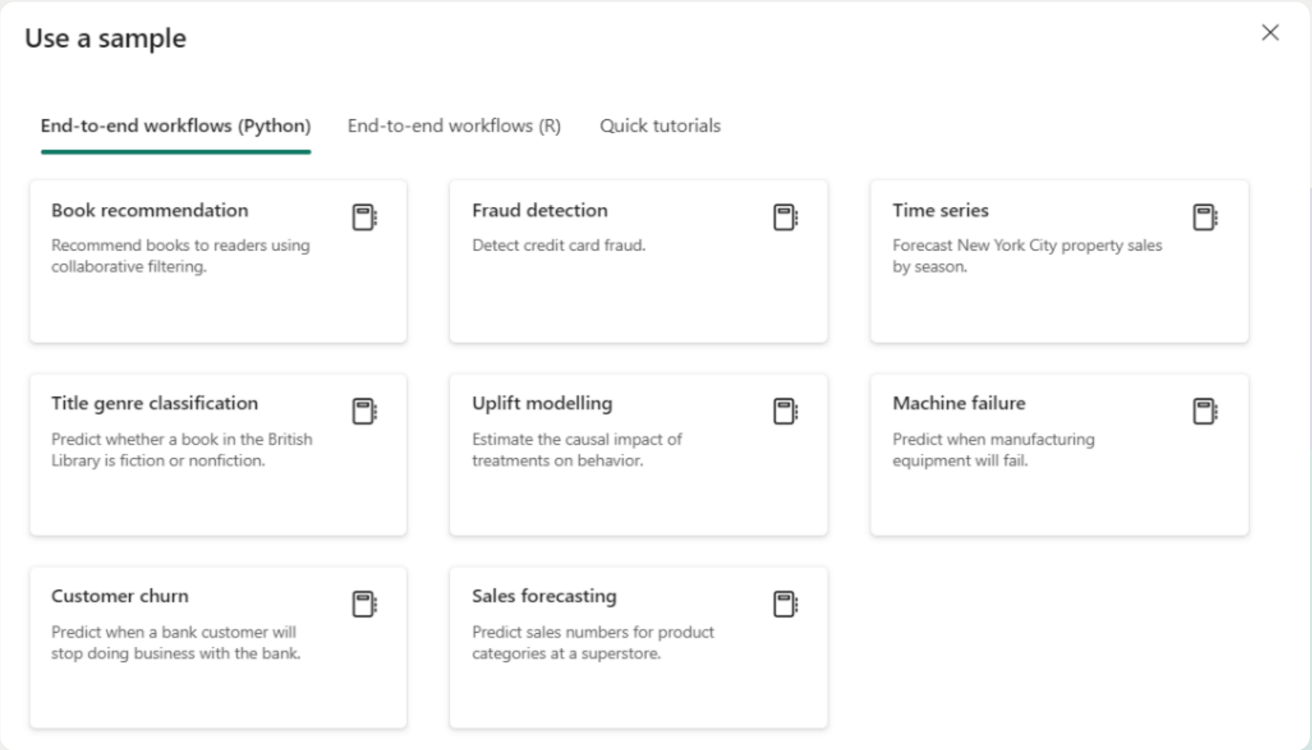


Get started with Data Science in Fabric

Leverage a rich set of data science samples for a variety of common use cases.

End-to-end examples of how to solve common problems with Machine Learning in MS Fabric

Newly added R samples and Quick tutorials



Machine learning with MLFlow

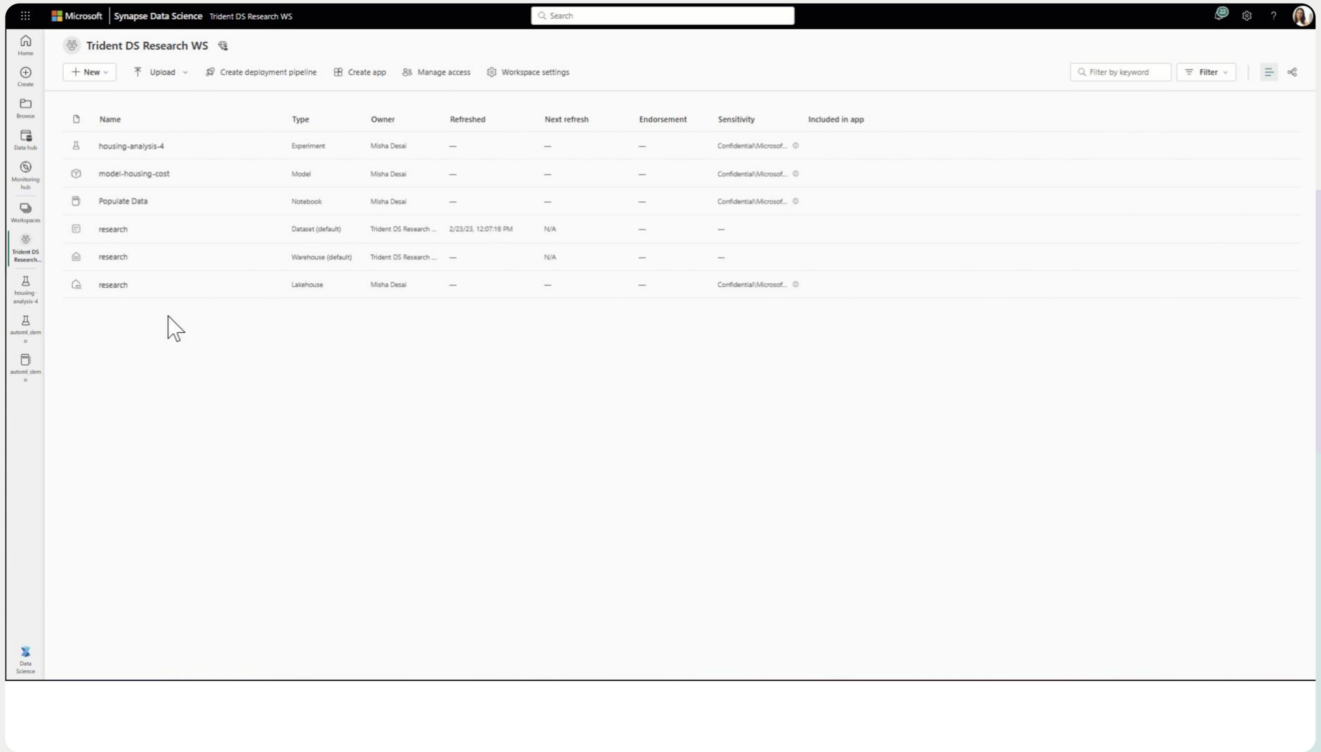
Built-in model & experiment tracking enables data scientists to track and compare their different experiment runs and model versions

Train, evaluate, and score machine learning models by using built-in Experiment and Model artifacts

---

Key Capabilities:

- Seamless integration with MLFlow for experiment tracking and model registration/deployment
- Automatically capture model metrics & parameters with built-in support for MLFlow auto-plogging
- ML integration with Azure OpenAI service makes it easier to use Apache Spark to process millions of prompts and scale workflow
- Model registry powered by AzureML



Model scoring

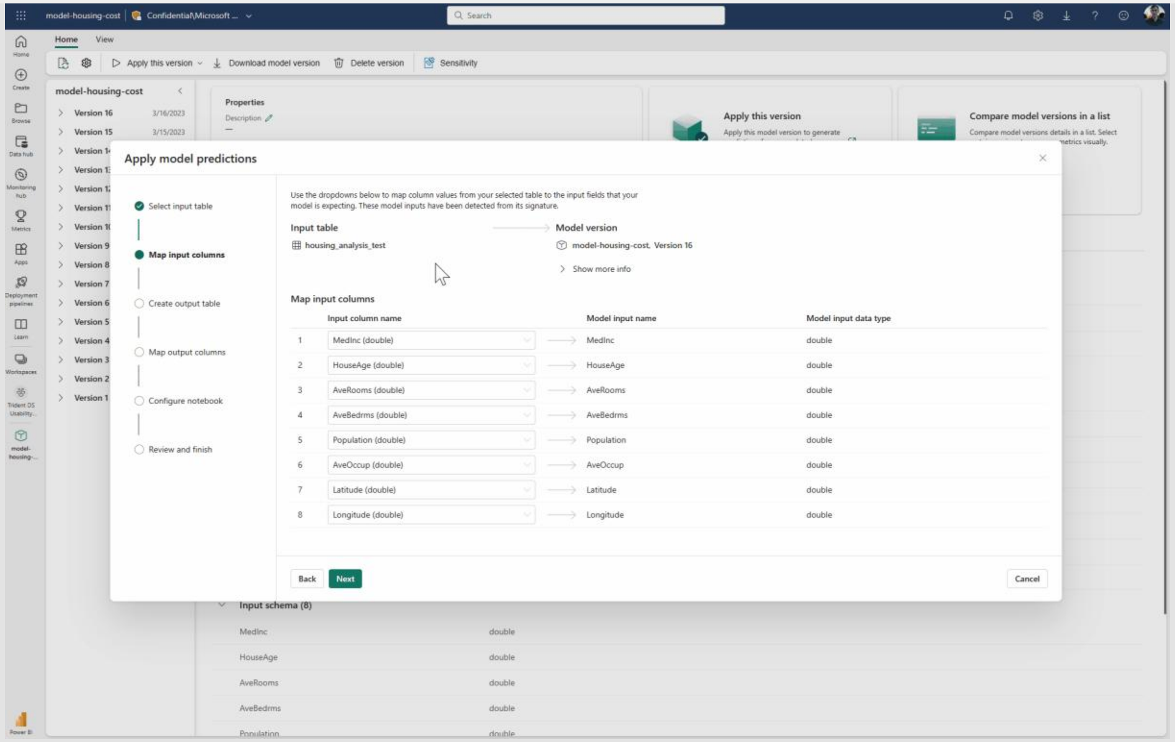
Easily enrich your data stored in the Lakehouse with machine learning predictions

Leverage Predict to score data at scale and write it back to your Fabric Lakehouse

---

Key Capabilities:

- Predict can be invoked from the secure boundaries of a Fabric notebook using APIs for maximum flexibility
- Leverage Predict directly from the UI using a guided experiment to deploy models in a matter of clicks
- Seamless integration with the Lakehouse, OneLake and Direct Lake meaning it is super simple to get your predictions into Power BI reports



Models and experiments with MLFlow

Built-in model & experiment tracking enables data scientists to track and compare their different experiment runs and model versions

---

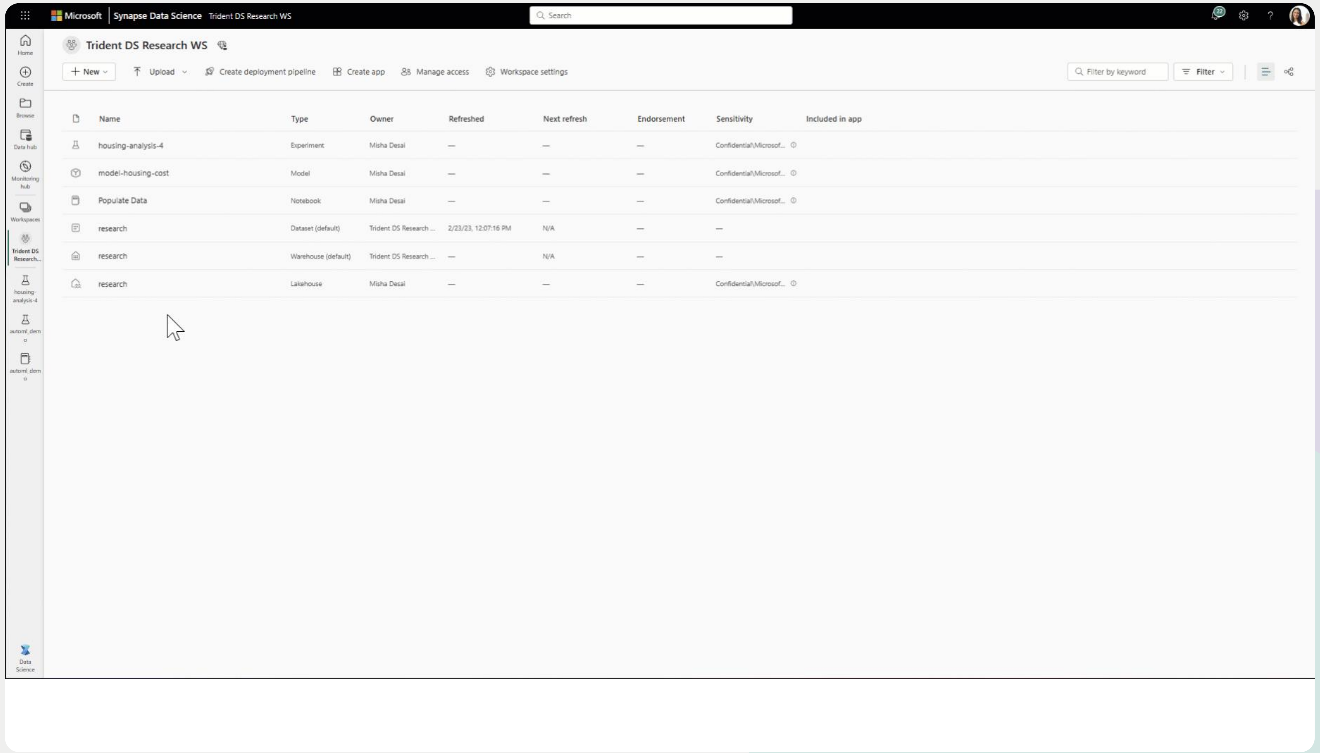
Automatically capture model metrics & parameters with built-in support for MLFlow auto-logging

---

Users can create and manage model and experiment items in Fabric

---

MLFlow model registry powered by Azure Machine Learning



Build your own powerful Gen AI experiences

Microsoft Fabric

Your unified solution for analytics

Enhance data insights

Build enterprise chat applications to uncover insights using natural language from structured, unstructured, and real-time data

Analyze customer interactions

Build speech analytics applications to enhance customer service, tailor support responses, and make data-driven decisions

Customize machine learning models

Train, deploy, and orchestrate machine learning models tailored to specific business needs—from predictive maintenance to customer sentiment analysis

Azure AI Foundry

Your gateway to AI apps and custom copilot experiences

OneLake serves as the connective tissue to build generative AI apps powered by your data

Azure AI Foundry Agent Service

Securely customize, orchestrate, and deploy AI agents

Model choice

Model choice and flexibility with the model catalog

Azure OpenAI Service

o1, o3-mini, GPT-4.1, 4o, etc

Models-as-a-Service

- Llama 3.1-405B-Instruct
- Mistral Large
- Cohere-Command-R-Plus

AI tools

Richest set of enterprise connectivity

Knowledge

Actions

Logic Apps · Azure Functions · OpenAPI · MCP

Trust

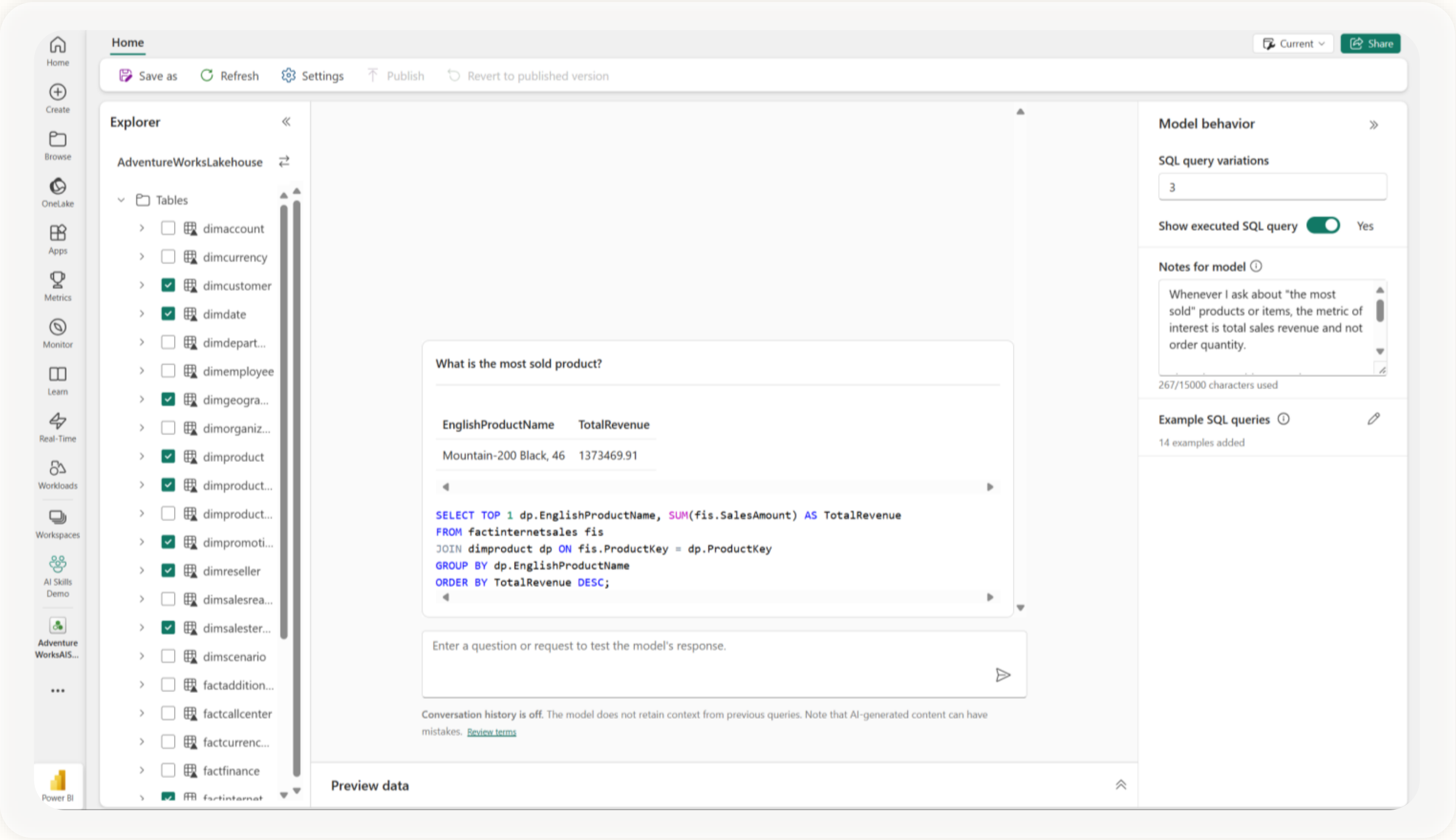
Customer control over data, networking, and security

- BYO-file storage
- BYO-search index
- BYO-virtual network
- BYO-thread storage
- OBO authentication
- Content filtering

Microsoft Fabric Data Agent

Deliver custom generative AI experiences for your data with Data Agents

- Allow your users to talk to data in Fabric
- Create, curate and configure a data expert on your data domain
- Share AI skills with other creators and consumers in Fabric



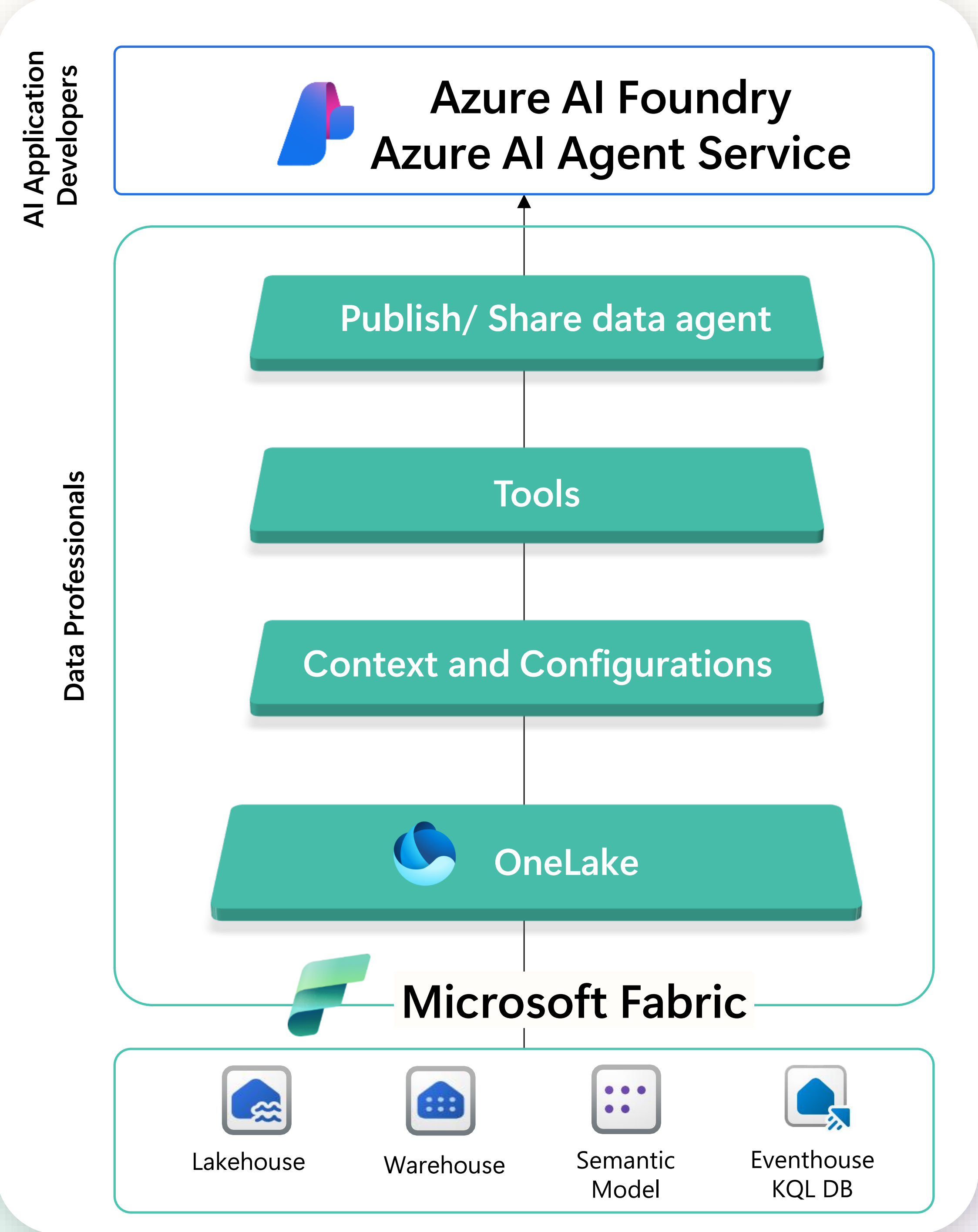
Microsoft Fabric Data Agent

Enhanced data interaction: create conversational agents that understand the org's unique data context, making it easy for non-technical users to get reliable, data-driven answers.

Faster time to insights: significantly reduce the time it takes to generate, explore data, and discover trends by automating the process of transforming natural language questions into queries.

Improved query capabilities: train the data agent with example instructions and queries, connect up to 5 data sources, including Lakehouse endpoints and KQL databases, and get full visibility into each step taken by the agent.

Access within or outside of Fabric.

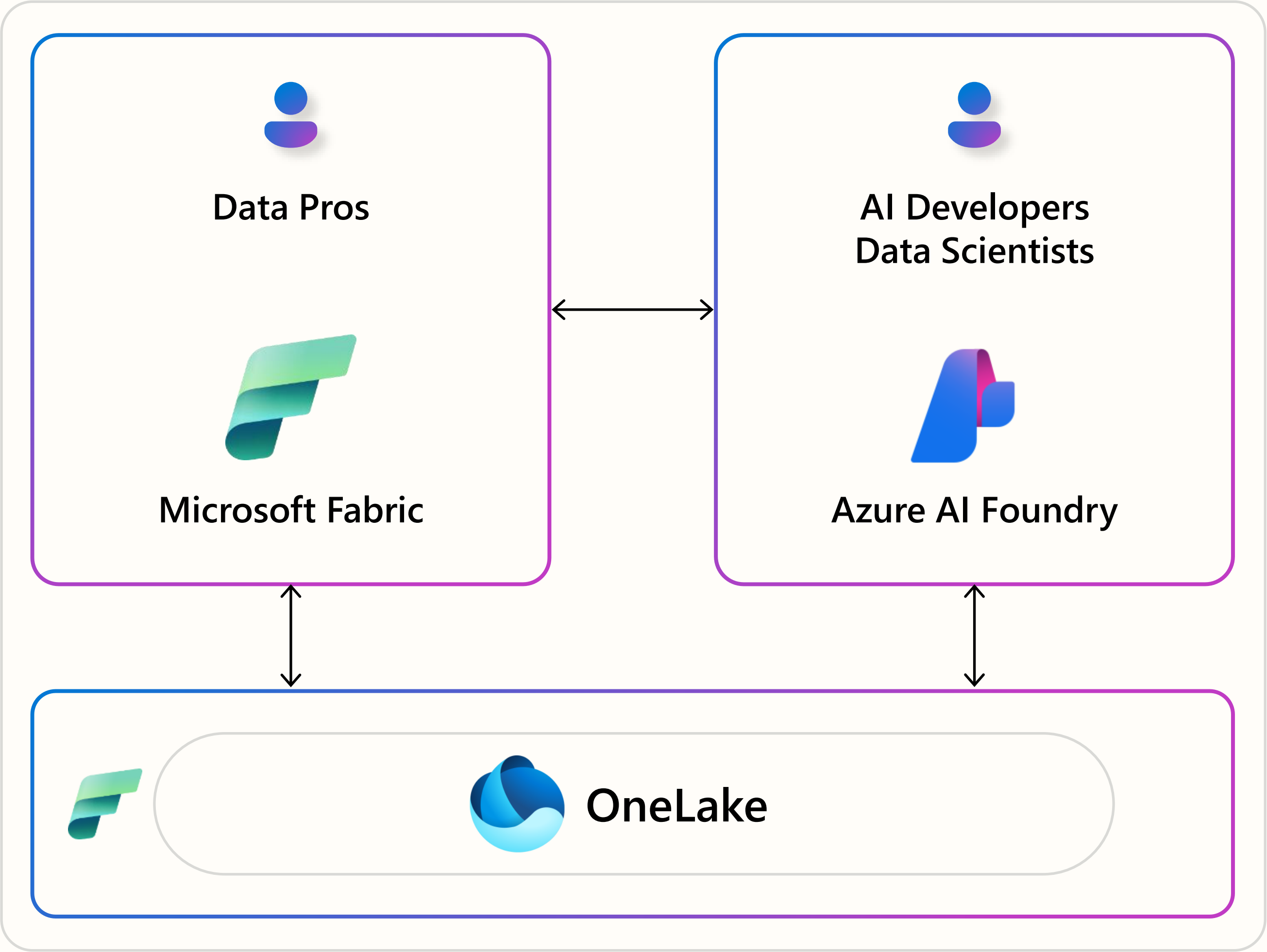


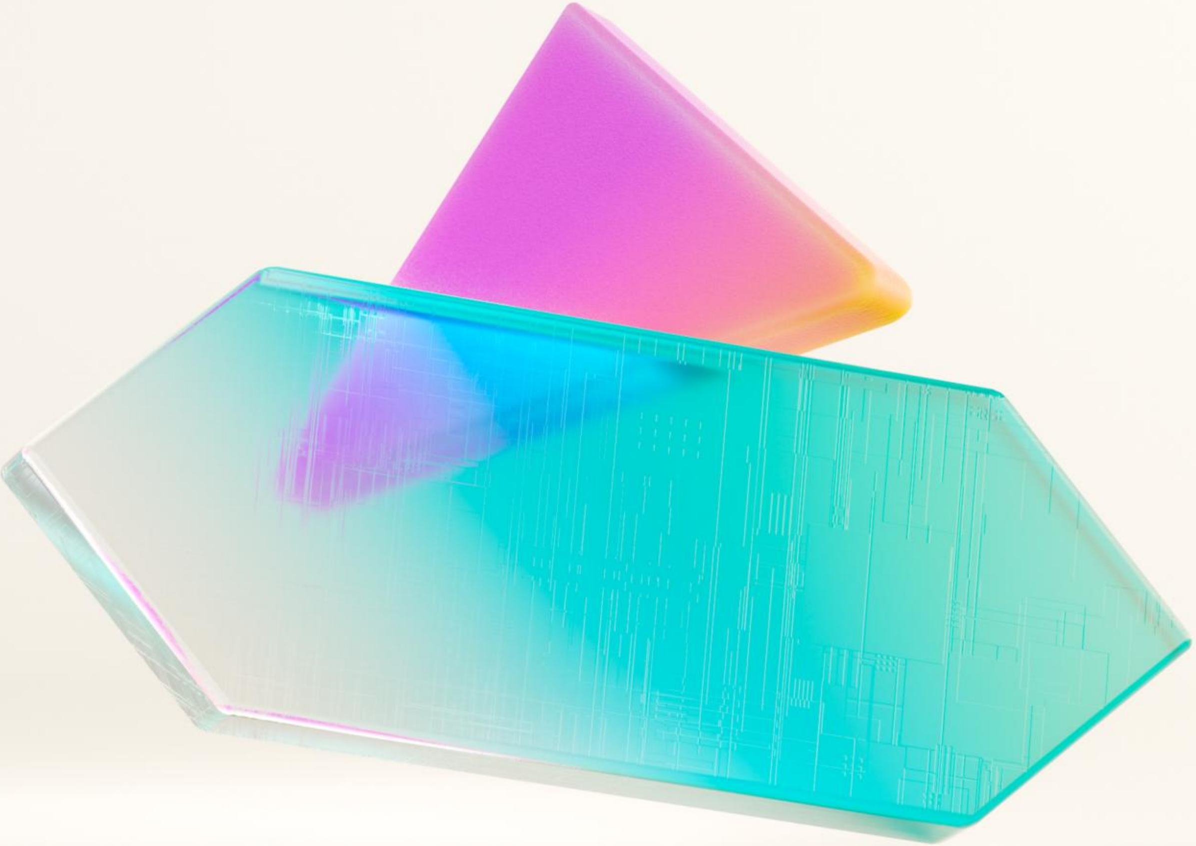
Microsoft Fabric and Azure AI Foundry integration

1. Data and AI Pros access, use and process data across the AI lifecycle with OneLake as the connective tissue.

2. Data Pros can leverage Azure AI Search to index and search data stored in OneLake, enabling developers to build intelligent AI apps with up-to-date insights.

3. Developers can leverage Fabric data agent integration in Azure AI Agent Service as a knowledge source to build enterprise AI agents that can uncover actionable insights from structured data.





You asked for more links?
Links you shall get!

Microsoft Fabric link library

Getting Started with Fabric

- What is Fabric?
- Fabric website
- Fabric trial
- Fabric licenses
- Buy a Fabric subscription
- Navigate the Fabric portal
- Workspaces in Fabric
- Fabric Learning Pathway
- Fabric Technical Documentation
- Fabric Guided Tour
- Fabric Industry Solutions
- Fabric Community
- See more

Copilot in Microsoft Fabric

- Copilot for Data Science and Data Engineering
- Copilot for Data Factory
- Copilot for Power BI

OneLake

- What is OneLake?
- What are shortcuts?
- Create a lakehouse with OneLake
- See more

Security, Governance, and Admin

- Fabric administration
- Data governance and compliance
- Security
- See more

End-to-End Tutorials

- Lakehouse tutorial
- Data Science tutorial
- Real-Time Intelligence tutorial
- Data Warehouse tutorial
- Power BI tutorial
- Data Factory tutorial

Other

- Azure Databricks trial

Microsoft Fabric workload link library

Data Factory

[What is Data Factory?]
[Create your first pipeline]
[Create your first dataflow]
[Connectors]
[See more]

Data Engineering

What is Data Engineering?
Create a Lakehouse
Create a Spark job definition
See more

Data Science

What is Data science?
Machine learning experiment
Use end-to-end AI samples
See more

Data Warehouse

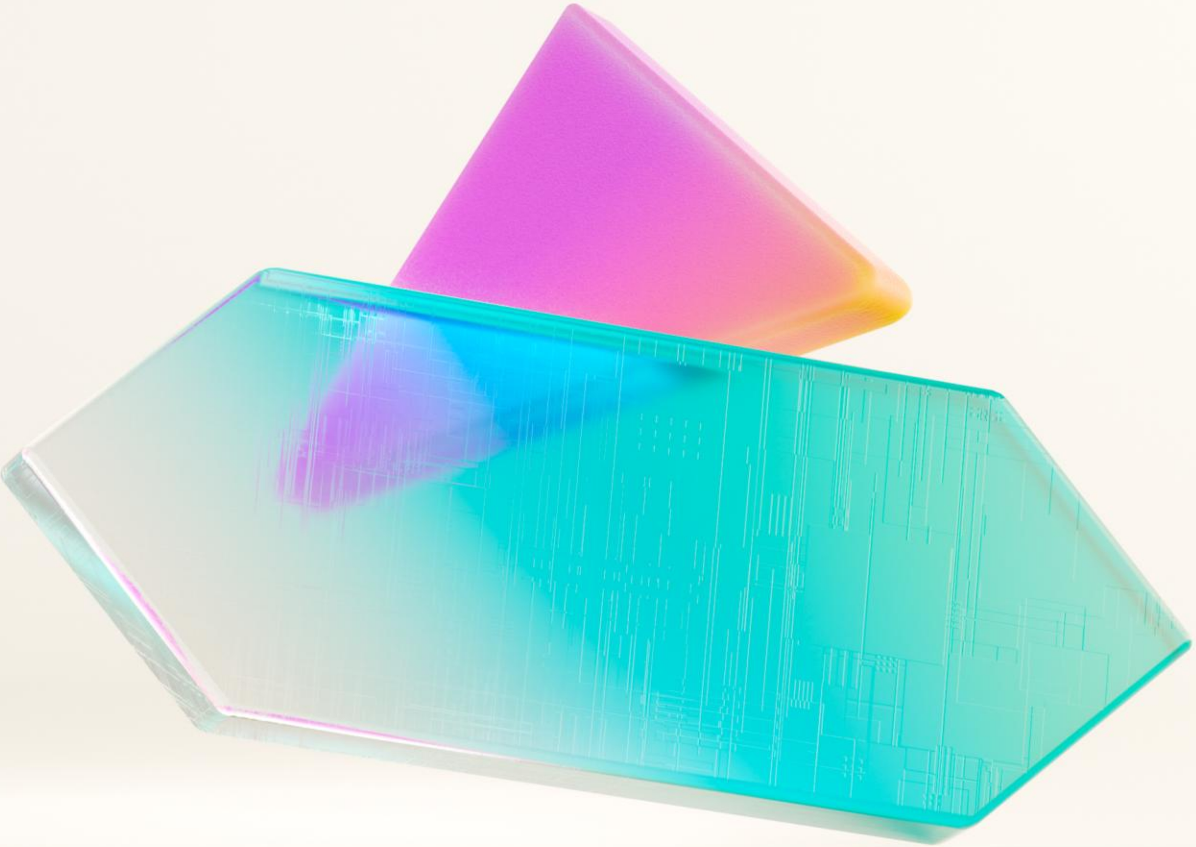
What is Data Warehouse?
Create a Warehouse
Query using SQL query editor
See more

Real-Time Intelligence

What is Real-Time Intelligence?
What is Event stream?
Create a database
See more

Power BI

Enable Microsoft Fabric for your organization
What is Power BI?
What is a datamart?
Azure and Power BI integration
See more



Community

Fabric Blog

Microsoft Fabric Updates Blog

Fabric October 2025 Feature Summary

AI · Announcements · Data Engineering · Data Factory · Data Science · Data Warehouse · Databases · Fabric platform · Lakehouse · Microsoft Fabric · Monthly Update · OneLake · Real-Time Intelligence

October 29, 2025 by Adam Saxton 65,672 Views

Share Tweet Like

This month's update delivers key advancements across Microsoft Fabric, including enhanced security with Outbound Access Protection and Workspace-Level Private Link, smarter data engineering features like Adaptive Target File Size, and new integrations such as Data Agent in Lakehouse. Together, these improvements streamline workflows and strengthen data governance for users.

Contents

Microsoft Fabric

Accelerate your data potential with a unified analytics solution that connects it all. Microsoft Fabric enables you to manage your data in one place with a suite of analytics experiences that seamlessly work together, all hosted on a lake-centric SaaS solution for simplicity and to maintain a single source of truth.

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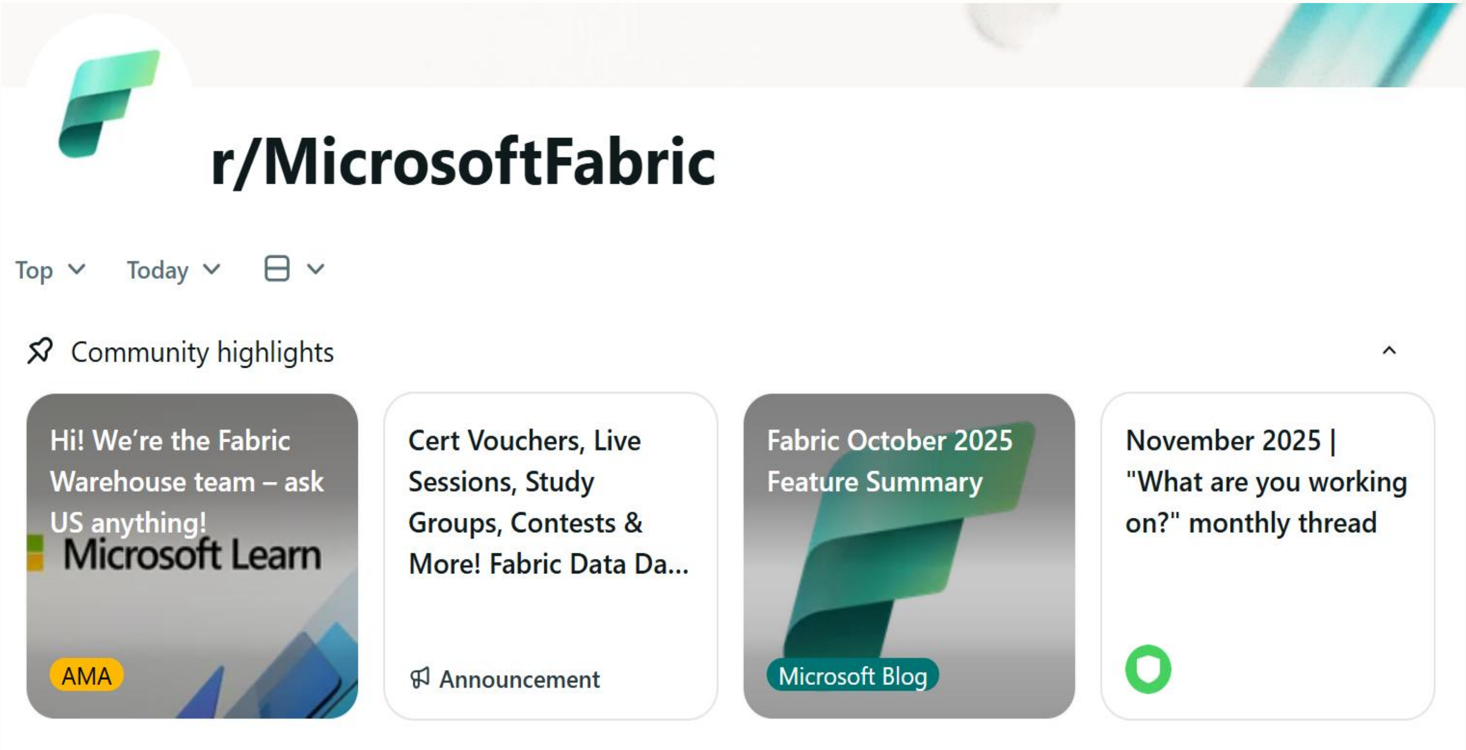
Fabric Subreddit

Exchange on ideas – ask the product group



Ever wanted to ask a question directly to the product group? Join our Ask Me Anything sessions.

Unsure how to solve a given problem? Ask the community. Let's harvest the experience of others.

Want to get visibility on latest development? Scan the reddit as well as our Fabric blogs.

Want to contribute to make Fabric a nice place to be? Share your story, demonstrate your skills, help others.

FabCon – 3 Days full of content data geeks and business will love

When

Each spring in the US and every autumn in Europe! Next stop is Atlanta in March and Barcelona in Septermber

Talk to the experts

Exchange with the community, customers but also directly with the people building Fabric.

Hands on?

Hands on! Each FabCon comes with a full day of hands on labs with community members or our product group members. Talk is cheap – lets get our hands dirty.